Highlights

- Accurate daily streamflow prediction in ungauged basins remains unresolved
- Transformer fuses 1 km meteorology and static basin data for daily streamflow prediction
- Outperforms baselines models in temporal accuracy and ungauged basin transfer
- First high-resolution cross-basin daily streamflow with static dynamic fusion

From Aggregated to Distributed: A Transformer Framework for Generalizable Streamflow Modeling Across Space and Time

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Abstract

Daily streamflow estimation in ungauged basins remains challenging because hydrologic response depends on spatially heterogeneous meteorological forcing and basin physiography. This study introduces **HydroTransformer**, a compact spatio-temporal Transformer architecture for distributed streamflow estimation. The model jointly encodes high-resolution Daymet meteorological forcings and spatially distributed static basin attributes. Dynamic inputs are processed through spatial patch encoding. Static rasters are then fused with dynamic representations using a Feature-wise Linear Modulation conditioning mechanism. A temporal Transformer then captures hydrologic memory across short, medium, and long context input sequences. HydroTransformer is evaluated on watersheds in Michigan's Lower Peninsula. The evaluation uses both temporal holdout and strict ungauged-basin spatial holdout settings. It is compared with Long Short-Term Memory (LSTM), Entity-Aware Long Short-Term Memory (EA-LSTM), Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM), and Informer baselines. Results show that HydroTransformer achieves strong and robust basin-wise performance, particularly when distributed static information is used. In the 365-day static-input setting, HydroTransformer obtains the highest median Nash-Sutcliffe Efficiency under both temporal and spatial splits. The median values are 0.670 and 0.668, respectively. These results demonstrate that explicit static-dynamic fusion of distributed basin information can improve streamflow estimation while using only six static spatial variables rather than large aggregated attribute sets. The proposed framework provides a practical step toward spatially informed deep learning for data-sparse and ungauged hydrologic prediction.

Keywords: Hydrology, Streamflow estimation, Spatio-temporal modeling,

1. Introduction

Streamflow, defined as the discharge of water moving through a river channel per unit time, is a foundational variable for water resources management [1]. Estimating streamflow in *ungauged* catchments, or basins without reliable discharge observations, is a long-standing challenge in hydrology, yet it is critical for reservoir operation, hydropower scheduling, flood-risk management, drought assessment, and ecological-flow protection. Such estimates enable forecast-informed operations that can enhance storage reliability and manage flood risk, and they support environmental flow targets crucial to the function of river ecosystems [2, 3]. Since the International Association of Hydrological Sciences decade on Prediction in Ungauged Basins, the hydrologic community has advanced a spectrum of regionalization approaches that transfer information from gauged to ungauged sites, ranging from statistical methods such as physiographical-space interpolation and top-kriging to parameter regionalization in conceptual and physically based models such as Soil and Water Assessment Tool and Hydrologic Engineering Center–Hydrologic Modeling System [4, 5, 6, 7]. Some studies aim to predict full continuous hydrographs at ungauged sites [8], whereas others directly regionalize hydrologic signatures such as mean flows, low flows, or peak-flow indices [9, 10]. Despite this progress, predictive skill remains uneven across hydroclimates and data-sparse regions. In particular, accurate daily streamflow estimation for ungauged basins remains a major unresolved challenge, underscoring the need for methods that are both accurate and transferable across heterogeneous catchments [11].

Traditional physically based rainfall–runoff models remain widely used because they encode hydrologic process knowledge and support scenario analysis. Models such as the Soil and Water Assessment Tool and the Hydrologic Engineering Center–Hydrologic Modeling System have been widely applied for watershed simulation and water-resource planning [12, 13]. However, their use in ungauged or data-sparse basins is limited by the need for extensive basin-specific inputs and careful parameter calibration. Additional limitations include uncertainty arising from equifinality in parameter estimation [14, 15]. These challenges become more pronounced when models are transferred across basins with different physiographic, climatic, and land-cover conditions [16]. Moreover, many traditional and operational workflows

represent a basin using lumped or overly aggregated forcings and attributes. Such representations can mask spatial heterogeneity in rainfall, snow accumulation, soil texture, topography, land cover, and storage processes. These masked factors influence runoff timing, magnitude, and recession behavior. Recent studies, therefore, emphasize explicit spatial representation and even hyper-resolution modeling to better capture distributed hydrologic processes [17, 18, 19]. These developments suggest that improving ungauged-basin prediction requires not only stronger temporal models but also architectures that can preserve spatially distributed forcing and basin-structure information.

In recent years, machine learning and deep learning have gained rapid traction in hydrology because they can learn rainfall–runoff relationships directly from data and reduce the need for labor-intensive basin-by-basin calibration [20, 21, 22, 23]. In large-sample settings, multi-basin deep learning models trained once and applied broadly have repeatedly outperformed regionally calibrated conceptual or physically based models. This performance highlights the promise of data-driven generalization across diverse hydrologic regimes [22, 23]. Among deep learning architectures, Long Short-Term Memory networks have emerged as a strong benchmark for streamflow prediction. They consistently deliver high performance across climates and even during extreme hydrologic conditions [24, 25, 26]. Entity-Aware Long Short-Term Memory models further improve regional learning by conditioning recurrent gates on static basin attributes [22]. Nevertheless, most recurrent approaches ingest basin-averaged meteorological time series and static attribute vectors. This limitation restricts their ability to exploit distributed forcing patterns and within-basin physiographic heterogeneity. To incorporate spatial information, convolutional and convolutional–recurrent models have been used to process gridded forcings or image-like hydrometeorological inputs [27, 28, 29]. These approaches can improve spatial feature extraction, but they often lack explicit mechanisms for fusing spatially distributed static rasters with time-varying meteorological fields. Furthermore, many lack robust transfer across contrasting ungauged basins. Recent guidance also cautions against single-basin training and emphasizes multi-basin learning as essential for reliability and transferability [30].

Transformer architectures use self-attention to capture long-range dependencies. These architectures have transformed the modeling of sequential and high-dimensional data in natural language processing, computer vision, and time-series forecasting [31, 32, 33]. Motivated by these strengths, hydrologic studies have begun exploring Transformer-based models for rainfall–runoff

and streamflow prediction. Some studies report gains with Temporal Fusion Transformers and specialized runoff Transformers. Conversely, others find performance comparable to strong Long Short-Term Memory baselines. These results depend on the dataset, input representation, and evaluation protocol [34, 35, 36, 37, 38]. In parallel, recent general-purpose time-series architectures such as PatchTST, iTransformer, and TimesNet have shown that temporal patching, variate-token attention, and multi-period representation learning can improve long-range forecasting in multivariate sequences [39, 40, 41]. However, these models are not designed specifically for hydrologic basins. In these basins, the predictors naturally organize as spatial fields over irregular watershed domains, and static physiographic structure strongly controls the dynamic runoff response. Recent hydrologic benchmarking further indicates that the relative advantage of recurrent and attention-based models is task-dependent. This reinforces the need to evaluate Transformer designs under hydrologically meaningful input representations and strict transfer settings [42].

Standardized large-sample datasets such as Catchment Attributes and Meteorology for Large-sample Studies and Caravan have been central to this progress because they provide long-term meteorological forcings, discharge observations, and basin attributes for broad regional learning [43, 44]. However, these datasets are commonly used through basin-averaged meteorological inputs and aggregated static descriptors. While such lumped representations are efficient and effective for many large-sample studies, they can obscure fine-scale hydro-meteorological variability. This variability is important for short-memory catchments, snow-influenced watersheds, convective storms, and extreme-flow events. Therefore, an important remaining gap is not simply the absence of deep learning models for streamflow prediction. The primary gap is the lack of architectures that jointly satisfy three requirements: preserving high-resolution spatial meteorological forcings, representing static basin properties as distributed spatial fields, and explicitly fusing static and dynamic information to support transfer to ungauged basins.

Consequently, this study addresses the open challenge of daily streamflow prediction using a hybrid Transformer-based framework that jointly encodes high-resolution meteorological forcings and spatially distributed static basin attributes. The proposed HydroTransformer uses spatial patch encoders to represent daily gridded meteorological fields and static basin rasters. It applies Feature-wise Linear Modulation to condition dynamic representations on basin-specific physiographic structure, and it employs a temporal

Transformer to model antecedent hydrologic memory. This design provides tangible advantages over prior approaches by retaining within-basin spatial heterogeneity and enabling explicit modulation of static and dynamic responses. It also reduces reliance on large sets of aggregated static attributes and supports evaluation under both future-year temporal holdout and strict ungauged-basin spatial holdout settings. The following specific objectives guide the study:

- **Spatially distributed static–dynamic fusion:** Rather than treating a basin as a single lumped unit, the model ingests spatially distributed meteorological and static basin patterns. This capability allows the same storm forcing to produce different learned runoff responses depending on basin-specific soil, topographic, and storage characteristics.
- **High-resolution forcing representation:** Rather than relying only on basin-aggregated meteorological inputs common in large-sample workflows, the model uses one-kilometer daily Daymet forcings [45]. This resolution better captures event timing, precipitation gradients, snow-related variability, and within-basin hydroclimatic structure.
- **Transferable ungauged-basin prediction:** The model is evaluated using both future-year temporal holdout and strict held-out-basin spatial validation. This approach allows an assessment of not only temporal robustness within observed basins but also transferability to basins with no discharge observations used during training.

2. Related Work

2.1. Classical statistical and regionalization approaches

Early daily streamflow estimation relied on statistical time-series and regionalization frameworks that transfer information from gauged to ungauged locations. Geostatistical methods, such as top-kriging, improved spatial interpolation of runoff indices and hydrographs by explicitly modeling spatial dependence across basins [6, 5]. Complementary regionalization strategies used similarity in climate and physiography to predict continuous flows or specific hydrologic indices at ungauged sites [8, 9]. These approaches established the importance of cross-basin information transfer, but their predictive

skill can be limited by assumptions of stationarity, manually defined similarity measures, and reduced ability to propagate fine-scale process heterogeneity into daily discharge estimates across diverse hydroclimatic settings.

2.2. Process-based hydrologic models

Process-based rainfall–runoff models remain the cornerstone for simulating the water balance because they encode hydrologic process understanding and support scenario analysis. Widely used frameworks include TOPMODEL [46], the Hydrologiska Byråns Vattenbalansavdelning model family [47], the Variable Infiltration Capacity land-surface model [48], the Soil and Water Assessment Tool for basin-scale applications [49], and the event/continuous Hydrologic Engineering Center–Hydrologic Modeling System [50]. These models provide physically interpretable simulations, but their deployment in ungauged or data-sparse basins remains difficult because they often require basin-specific calibration, detailed parameterization, and reliable forcing and physiographic inputs. Parameter equifinality and transferability limitations can further reduce confidence when calibrated parameters are moved from gauged to ungauged catchments.

2.3. Data-driven streamflow models: recurrent neural networks and beyond

With the rise of large-sample hydrology, deep learning has achieved strong accuracy for regional streamflow prediction [24, 22]. Long Short-Term Memory networks have become a widely used benchmark because they can learn nonlinear rainfall–runoff relationships and retain antecedent hydrologic information over daily to seasonal time scales [24]. Subsequent studies have extended this line of work to multi-timescale prediction [51], broad climatic evaluation [26], and extreme-flow prediction [25]. These gains have been supported by standardized datasets such as Catchment Attributes and Meteorology for Large-sample Studies [43] and Caravan [44], which provide basin attributes, meteorological forcings, and discharge observations for large-sample regional modeling.

Despite their strong performance, most recurrent neural network models represent each basin using basin-averaged meteorological time series and static attribute vectors. This lumped representation is efficient and has proven effective in many regional streamflow settings, but it limits the model's ability to exploit spatial heterogeneity in precipitation, snow, soils, topography, and land cover. Entity-Aware Long Short-Term Memory models partially address this limitation by conditioning recurrent gates on static basin

attributes [22], but the static information is typically represented as aggregated catchment descriptors rather than spatially distributed raster fields. Therefore, recurrent models provide strong temporal baselines, but they do not fully address the need for spatially explicit static–dynamic hydrologic fusion.

2.4. Deep spatio-temporal models and fusion of static attributes

Recent work has moved beyond lumped inputs by learning from gridded forcings, spatial connectivity, and attention-based representations. Convolutional–recurrent models, including Convolutional Long Short-Term Memory architectures, ingest spatial meteorological data to improve regional prediction and enable spatial interpretation [27]. Related Convolutional Neural Network–Long Short-Term Memory models have also been explored for streamflow and hydrologic prediction [28, 29]. These models can learn local spatial patterns from gridded inputs, but many studies either omit static basin rasters or incorporate static information only indirectly. As a result, their ability to explicitly condition dynamic runoff response on spatially distributed physiographic properties remains limited.

Graph-based hydrologic models offer another route for incorporating spatial structure. Physics-guided graph neural networks can encode river-network connectivity, basin attributes, and upstream–downstream dependencies [52]. Such models are well suited for representing hydrologic networks and routing relationships, but they do not directly solve the problem of fusing high-resolution gridded meteorological fields with static raster layers over individual basin domains.

Transformer-based models have also been introduced for rainfall–runoff and streamflow prediction. Specialized runoff Transformers and models for prediction in newly gauged or low-data basins have demonstrated competitive or superior performance relative to recurrent baselines in some settings [35, 36]. Temporal Fusion Transformers combine recurrent processing, gating, static covariate encoding, and attention, and have recently shown benefits for streamflow forecasting [53, 34]. However, most hydrologic Transformer studies still operate on basin-averaged multivariate time series and aggregated static attributes. Therefore, they improve long-range temporal modeling but do not fully exploit spatially distributed dynamic forcings and static basin rasters.

In parallel, recent general-purpose time-series architectures like PatchTST, iTransformer, Informer, and TimesNet have advanced long-range multivari-

ate forecasting through temporal patching, variate-token attention, and multi-period temporal representation learning [39, 40, 41, 38]. These methods are useful modern baselines for sequence modeling, but they are not designed around irregular watershed domains, gridded meteorological forcings, or static physiographic rasters. Recent hydrologic benchmarking further indicates that the relative advantage of recurrent and attention-based architectures is task-dependent, reinforcing the need to evaluate model designs under hydrologically meaningful input representations and transfer settings [42].

Table 1: Structured comparison of representative streamflow and time-series modeling approaches.

Method family	**Studies**	**Input representation**	**Fusion setting**	**Key limitation**
Statistical regionalization	[6, 5, 8, 9]	Hydrologic signatures or aggregated hydrographs	Similarity transfer; ungauged transfer	Limited nonlinear dynamics and spatial heterogeneity.
Process-based models	[46, 47, 48, 49, 50]	Meteorological forcings and physical parameters	Process equations; calibrated simulation	Calibration and parameter transfer are difficult.
Large-sample recurrent models	[24, 22, 25, 26, 30]	Basin-averaged meteorological time series	Recurrent sequence learning; multi-basin training	Lumped inputs limit spatial process representation.
Entity-aware recurrent models	[22, 25]	Basin-averaged forcings and static vectors	Static gating in recurrent networks	Static information remains aggregated.
Convolutional–recurrent models	[27, 28, 29]	Gridded or image-like dynamic forcings	Convolution plus recurrence	Distributed static-raster fusion is limited.
Graph-based hydrologic models	[52]	River-network or basin time series	Graph message passing	Focuses on network structure, not raster fusion.
Hydrologic Transformer models	[34, 35, 36, 37]	Mostly basin-averaged multivariate time series	Attention or gating; temporal/spatial transfer	Usually relies on lumped forcings and attributes.
General time-series Transformers	[39, 40, 41, 38]	Multivariate time-series inputs	Temporal patching or variate-token attention	Not designed for watershed rasters.
Hydrologic architecture benchmarking	[42]	Multi-source hydrologic time series	Recurrent or attention-based fusion	Does not address distributed raster fusion.
Proposed HydroTransformer	**This study**	**Gridded daily meteorology and distributed static basin rasters**	**Spatial patch encoding, Feature-wise Linear Modulation, and temporal Transformer**	**Jointly models distributed forcings, static rasters, and ungauged transfer.**

Note: Lumped inputs refer to basin-averaged predictors or attribute vectors, whereas distributed inputs refer to gridded or raster-based spatial representations.

Based on the structured comparison in Table 1, existing approaches typically address only part of the streamflow prediction problem. Statistical regionalization and process-based models provide hydrologically meaningful transfer mechanisms, but they require strong assumptions, calibration, or basin-specific parameterization. Large-sample recurrent models and Entity-Aware Long Short-Term Memory networks provide strong temporal baselines, but they generally rely on basin-averaged forcings and aggregated static attributes. Convolutional–recurrent and graph-based models introduce spatial or network structure, yet they do not explicitly fuse distributed meteorological forcings with distributed static basin rasters. Transformer-based hydrologic models and recent general-purpose time-series architectures improve long-range temporal modeling, but most remain designed around lumped multivariate sequences rather than irregular watershed domains with spatially distributed predictors. In contrast, HydroTransformer jointly encodes high-resolution distributed meteorological forcings and spatially distributed static basin rasters, fuses them through Feature-wise Linear Modulation, and evaluates transferability under both temporal holdout and strict ungauged-basin holdout settings. To the best of our knowledge, this combination of distributed static–dynamic fusion and ungauged-basin evaluation has not been explicitly addressed in prior streamflow modeling studies.

3. Methodology

This section describes the study region, data sources, baseline models, and the proposed HydroTransformer architecture. Our objective is to develop a spatio-temporal DL framework capable of accurately estimating daily streamflow across heterogeneous watersheds using high-resolution spatially distributed meteorological forcings and physical basin attributes. The following subsections detail the region of study, the construction of the dataset, baseline models, and the proposed architecture.

3.1. Region of Study

This study focused on 59 watersheds located in Michigan's Lower Peninsula, United States. These watersheds were selected based on three criteria: (i) availability of complete daily meteorological records, (ii) availability of continuous streamflow observations, and (iii) coverage of the 2000–2020 period without significant data gaps. The region provides an appropriate

testbed for evaluating streamflow modeling and ungauged-basin transfer under Great Lakes hydrologic conditions. The study watersheds represent a range of basin sizes, land-cover conditions, soil properties, topographic gradients, drainage characteristics, and hydrologic response behaviors. They include mixed agricultural, forested, and urban landscapes, as well as areas influenced by glacially derived landforms, variable groundwater contributions, snowmelt processes, and lake-effect precipitation [54, 55, 56].

Although Michigan's Lower Peninsula is geographically bounded, it is not hydrologically homogeneous. Watershed response varies substantially due to differences in geology, soil properties, slope, land use, drainage characteristics, groundwater contribution, snow influence, and climatic gradients. As a result, the selected watersheds provide meaningful variability for evaluating regional model transferability. In particular, northern Lower Peninsula watersheds are commonly underlain by coarse glacial deposits that promote infiltration and groundwater storage, producing relatively stable, groundwater-supported flows.

In contrast, parts of southern and southeastern Michigan contain finer soils, greater agricultural and urban influence, and drainage modifications. This results in more runoff-responsive streamflow behavior, including sharper rises and recessions following storm events [56]. Consistent with this heterogeneity, hydrograph-separation analysis across the United States portion of the Great Lakes Basin shows that the groundwater component of streamflow varies widely among streams. This component ranges from 25% to 97%, with higher groundwater contributions associated with coarse-textured soils and less disturbed vegetation [57].

Therefore, while the study area does not represent the full hydroclimatic diversity of the United States or global watersheds, it contains substantial variability in hydrologic response characteristics. It also provides a meaningful regional testbed for evaluating temporal generalization and transferability to unseen basins. Accordingly, model generalization in this study should be interpreted as regional generalization within the Michigan Lower Peninsula rather than universal transferability across all hydroclimatic regimes. This distinction is important because artificial intelligence hydrologic models are inherently domain-dependent. Their transferability is strongly influenced by watershed-specific controls such as climate, soils, geology, topography, land use, groundwater contribution, and human modification. The spatial-holdout experiment was therefore designed to evaluate transferability to unseen watersheds within this heterogeneous regional domain.

3.2. Data

Dynamic meteorological forcings, static basin attributes, and observed streamflow records are integrated to construct an end-to-end dataset for training and evaluation of the proposed model.

3.2.1. Data Sources

Daily gridded meteorological inputs are obtained from Daymet [58], which provides 1 km $\times$ 1 km resolution fields across the United States. For each watershed, Daymet grids are clipped to the basin polygon, and the following dynamic variables are extracted: precipitation (mm day^{-1}), minimum temperature (°C), maximum temperature (°C), shortwave radiation (W m^{-2}), vapor pressure (Pa), daylength (s), and snow water equivalent (mm). These variables supply the model with high-resolution, spatially explicit meteorological drivers that capture event timing, intensity, and within-basin heterogeneity. Static physical and physiographic descriptors are derived from HydroAtlas [59], a globally standardized hydrologic attribute database based on HydroSHEDS. A total of 153 basin attributes, aggregated over the basin area, are used to run the baseline experiments. But for each watershed, raster layers representing the digital elevation model (DEM), soil properties (clay, sand, and silt), available water content (AWC), and field capacity (FC) are extracted. These six static variables were selected because they represent major physiographic controls on runoff generation while remaining computationally tractable as distributed raster inputs. DEM provides information about topographic gradients and potential runoff pathways. Clay, sand, and silt represent soil texture, which controls infiltration, drainage, and water retention. Available water content and field capacity describe soil-water storage capacity and therefore influence antecedent wetness, baseflow support, and runoff response. Compared with the 153 aggregated attributes used by the LSTM-based baselines, this compact raster set allows HydroTransformer to preserve spatially distributed basin structure without greatly increasing input dimensionality or memory cost. These static attributes are resampled to the Daymet grid resolution (1 km) to ensure spatial alignment with the dynamic variables. Treating static variables as rasters allows the model to capture meaningful spatial gradients such as slope transitions, soil heterogeneity, and geomorphic structure. Ground-truth discharge values are obtained from the USGS [60] database. Daily mean streamflow (m^3 s^{-1}) is extracted for each gauged basin and aligned temporally with the Daymet forcings. Only

basins with reliable, full-record observations over the 2000–2020 period are retained.

3.2.2. Data Pre-processing

Before model training, all input data (dynamic forcings, static attributes, and streamflow observations) underwent a standardized pre-processing pipeline to ensure consistency, remove artefacts, and facilitate the learning of hydrologically meaningful relationships.

Meteorological forcings from Daymet V4 and static basin attributes from HydroATLAS were provided initially as NetCDF rasters. For each of the 59 selected watersheds, the rasters were clipped precisely to the watershed polygon, and the resulting per-basin grids were saved in HDF5 (.h5) format for efficient downstream access. All input rasters were scanned for missing values (NaNs) or infinite values. Any occurrence of NaN or ± (e.g., masked grid cells outside the domain) was set to 0.0 to prevent the propagation of undefined values through the network. For discharge data, basins or time series exhibiting significant daily data gaps were excluded. The remaining daily mean discharge Q_t was area-normalized (to account for varying watershed size) and then a natural logarithm transformation was applied:

$$Q'_t = \ln\big(Q_t/\text{area}\big).$$

Precipitation and SWE, both variables are strictly non-negative and exhibit strong right-skew; accordingly, a logarithmic transformation is applied:

$$X' = \ln(X + \varepsilon),$$

where ε is a small constant for numerical stability. Other climate variables, such as minimum/maximum temperature, shortwave radiation, vapor pressure, and daylength, were standardized (zero mean, unit variance). The mean and variance statistics were computed using only the training set, and the same transformation was applied to the validation/test subsets to avoid information leakage. For static variables, raster layers (DEM, soils, available water content, field capacity) were used in their raw form (no transformation), preserving physically interpretable units and within-basin gradients. Table 2 provides an overview of all the processing steps performed on the data. Each basin's dynamic input sequence and discharge target were aligned by date over the period 2000–2020. Any leap day (February 29) was handled by removing the last day of the year (December 31). Input sequences of

length L days were constructed to correspond to the output streamflow at the final day of the window.

Variable group	**Pre-processing steps**
Discharge (raw)	1. Remove missing rows and basins with no data. 2. Normalize by basin area. 3. Apply log-transform.
Precipitation, SWE	Log-transform.
Dynamic climate variables	Standardize (global training-set μ and σ)
Static catchment attributes	None (raw features)

Table 2: Pre-processing overview for model input variables.

3.3. Baseline Models

To provide a rigorous benchmark for the proposed architecture, four deep learning baselines are implemented and compared. These include Long Short-Term Memory (LSTM), Entity-Aware Long Short-Term Memory (EA–LSTM), a Convolutional Neural Network–Long Short-Term Memory hybrid (CNN–LSTM), and Informer. These models were selected to represent the main sequence-modeling and static–dynamic fusion paradigms used in hydrologic deep learning.

The LSTM baseline represents the standard recurrent sequence-modeling approach. It can learn long-term temporal dependencies from meteorological forcing sequences [61, 24]. When static attributes are used, they are incorporated as vector inputs. This represents the common concatenation-based fusion strategy. The EA–LSTM baseline explicitly incorporates static basin attributes into the input-gate mechanism. This represents the static-gating paradigm for basin-specific hydrologic response modeling [22]. The CNN–LSTM hybrid first extracts convolutional features from dynamic inputs. It then models temporal dependencies through an LSTM. This provides a baseline that combines spatial feature extraction with recurrent temporal modeling [27]. Informer is included as a Transformer-based sequence-modeling baseline. It is used to compare HydroTransformer with a recent attention-based architecture for long-sequence time-series forecasting [38].

Together, these baselines cover the dominant alternatives to the proposed design. These alternatives include concatenation-based static fusion, static gating in recurrent networks, convolutional spatial encoding followed by recurrence, and attention-based temporal modeling. All models are evaluated using the same meteorological inputs, sequence lengths, temporal and spatial holdout splits, preprocessing procedure, and evaluation metrics. Static attributes are provided when supported by the corresponding architecture. HydroTransformer differs from these baselines by explicitly encoding gridded meteorological forcings and distributed static basin rasters. It fuses them through spatial patch encoding and Feature-wise Linear Modulation.

3.4. Proposed Architecture

A unified spatio-temporal Transformer architecture is proposed that fuses gridded meteorological forcings and static basin attributes through a conditioning mechanism, enabling spatial heterogeneity within each basin and basin-specific response modulation. The architecture comprises three key stages: (i) spatial encoding of dynamic forcings, (ii) static attribute encoding and Feature-wise Linear Modulation (FiLM)-based conditioning, and (iii) temporal encoding and discharge prediction. The detailed version of architecture is presented in Figure 1 and explained in Algorithm 1. Table 3 provides the input and output dimensions of each layer in proposed architecture.

The patch size was selected to balance spatial detail and computational feasibility. Because the meteorological forcings and static rasters are aligned to the 1 km Daymet grid, a patch size of 8 corresponds approximately to an 8 km x 8 km spatial neighborhood. This resolution is sufficient to preserve within-basin heterogeneity in precipitation, snow, soil, and topographic patterns while reducing the number of spatial tokens passed to the Transformer. Smaller patches increase token count and attention cost, whereas larger patches overly smooth local hydrologic variability. Thus, $P = 8$ provides a practical compromise between spatial resolution, memory demand, and representation capacity for the selected watershed grids.

3.4.1. Spatial encoding of dynamic forcings

Let the input tensor of dynamic meteorological variables be

$$X \in \mathbb{R}^{B \times L \times C_{\text{dyn}} \times H \times W},$$

where B is the batch size, L the temporal context length, C_{dyn} the number of dynamic channels, and (H, W) the spatial grid resolution. For each day

Algorithm 1 Forward pass of Vision Transformer ViT Temporal Fusion model (FiLM mode)

Require: Dynamic input $X \in \mathbb{R}^{B\times L\times C_{\text{dyn}}\times H\times W}$, Static rasters $S \in \mathbb{R}^{B\times C_{\text{stat}}\times H_s\times W_s}$

Ensure: Predicted discharge $\hat{y} \in \mathbb{R}^{B\times 1}$

$X \leftarrow \texttt{sanitize}(X)$
$S \leftarrow \texttt{resize_static}(S,\, H,\, W)$
$S \leftarrow \texttt{sanitize}(S)$
for $t = 1$ to L **do**
    $(Z_t^{\text{dyn}}, H_p, W_p) \leftarrow \texttt{PatchEmbed2D}(X_{:,t,:,:})$
    $z_t \leftarrow \texttt{SpatialVit}(Z_t^{\text{dyn}}, H_p, W_p)$ ▷ mean-pool to $\mathbb{R}^{B\times D}$
end for
Stack: $Z \leftarrow [z_1, \ldots, z_L] \in \mathbb{R}^{B\times L\times D}$
$(Z^{\text{stat}}, H_{pS}, W_{pS}) \leftarrow \texttt{PatchEmbed2D}(S)$
$s \leftarrow \texttt{SpatialVit}(Z^{\text{stat}}, H_{pS}, W_{pS})$ ▷ $\mathbb{R}^{B\times D}$
$s \leftarrow \texttt{LayerNorm}(s)$
$(\gamma, \beta) \leftarrow \texttt{MLP_FiLM}(s)$ ▷ $\gamma, \beta \in \mathbb{R}^{B\times D}$
$\gamma \leftarrow 0.1 \times \tanh(\gamma)$
for $t = 1$ to L **do**
    $\tilde{z}_t \leftarrow z_t \odot (1 + \gamma) + \beta$
end for
$\tilde{Z} \leftarrow [\tilde{z}_1, \ldots, \tilde{z}_L] \in \mathbb{R}^{B\times L\times D}$
$\tilde{Z} \leftarrow \tilde{Z} + E^{\text{1D}}$ ▷ 1-D temporal positional encoding
if causal = True **then**
    $\hat{H} \leftarrow \texttt{TemporalVit}(\tilde{Z},\ \texttt{causal_mask} = True)$
else
    $\hat{H} \leftarrow \texttt{TemporalVit}(\tilde{Z})$
end if
$h \leftarrow \hat{H}_{:,-1,:}$ ▷ last time-step embedding, shape $\mathbb{R}^{B\times D}$
$\hat{y} \leftarrow \texttt{Head}(h)$ ▷ regression head
return $\hat{y}$

Table 3: Input and output dimensions at each stage of the proposed HydroTransformer architecture.

Model stage	**Layers**	**Input dimension**	**Output dimension**
Dynamic input	N/A	$B \times L \times C_{\text{dyn}} \times H \times W$	$B \times L \times C_{\text{dyn}} \times H \times W$
Static input resizing	N/A	$B \times C_{\text{stat}} \times H_s \times W_s$	$B \times C_{\text{stat}} \times H \times W$
Dynamic patch embedding	1	$(BL) \times C_{\text{dyn}} \times H \times W$	$(BL) \times N \times D$
Static patch embedding	1	$B \times C_{\text{stat}} \times H \times W$	$B \times N \times D$
Spatial Transformer for dynamic inputs	4	$(BL) \times N \times D$	$(BL) \times N \times D$
Dynamic mean pooling	1	$(BL) \times N \times D$	$B \times L \times D$
Spatial Transformer for static inputs	4	$B \times N \times D$	$B \times N \times D$
Static mean pooling and normalization	1	$B \times N \times D$	$B \times D$
Feature-wise Linear Modulation generator	1	$B \times D$	$B \times 2D$
Static–dynamic conditioning	1	$B \times L \times D$	$B \times L \times D$
Temporal Transformer encoder	6	$B \times L \times D$	$B \times L \times D$
Temporal readout	1	$B \times L \times D$	$B \times D$
Regression head	2	$B \times D$	$B \times H_{\text{out}}$

Here, B is the batch size, L is the input sequence length, C_{dyn} is the number of dynamic meteorological variables, C_{stat} is the number of static raster variables, H and W are the spatial grid dimensions, H_s and W_s are the original static raster dimensions, P is the patch size, $N = (H/P)(W/P)$ is the number of patches, D is the embedding dimension, and H_{out} is the prediction horizon. In this study, $P = 8$, $D = 192$, and $H_{\text{out}} = 1$.

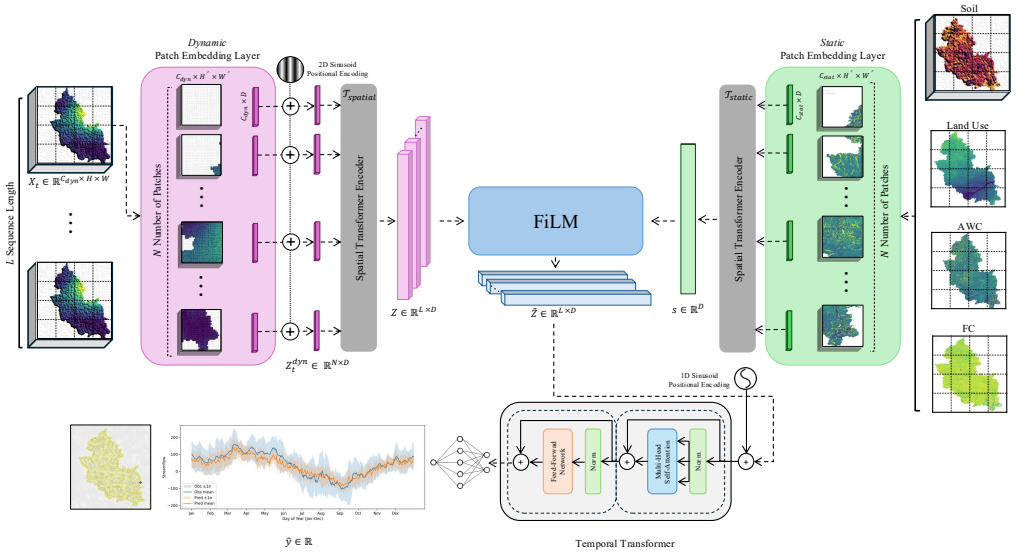


Figure 1: Dynamic meteorological forcings X_t are patchified and encoded using a shared spatial Transformer. Static basin rasters are encoded similarly and pooled to form a static embedding, which conditions dynamic features via FiLM modulation. The conditioned sequence is processed by a temporal Transformer, and the final representation is mapped to daily streamflow through a regression head.

$t \in \{1, \ldots, L\}$, the daily frame $X_{:,t,:,:}$ is patch-embedded via a 2-D convolution into

$$Z_t^{\text{dyn}} = \text{PatchEmbed}(X_{:,t,:,:}) \in \mathbb{R}^{B \times N \times D},$$

where $N = \frac{H}{P} \cdot \frac{W}{P}$ (patch count) and D is the embedding dimension. A fixed 2-D sinusoidal positional encoding $E^{\text{2D}} \in \mathbb{R}^{N \times D}$ is added yielding

$$\tilde{Z}_t^{\text{dyn}} = Z_t^{\text{dyn}} + E^{\text{2D}}. \tag{1}$$

These tokens pass through a shared spatial Transformer encoder $\mathcal{T}_{\text{spatial}}$, giving

$$U_t^{\text{dyn}} = \mathcal{T}_{\text{spatial}}(\tilde{Z}_t^{\text{dyn}}) \in \mathbb{R}^{B \times N \times D}. \tag{2}$$

Mean pooling is applied across patches to obtain the per-day embedding

$$z_t = \frac{1}{N} \sum_{i=1}^{N} U_{t,:,i}^{\text{dyn}} \in \mathbb{R}^{B \times D}. \tag{3}$$

Stacking yields the sequence

$$Z = [\, z_1, z_2, \ldots, z_L] \in \mathbb{R}^{B \times L \times D}. \tag{4}$$

3.4.2. Static attribute encoding

Static attributes are provided as rasters

$$S \in \mathbb{R}^{B \times C_{\text{stat}} \times H_s \times W_s},$$

resampled to (H, W). After patch embedding and positional encoding, tokens are

$$Z^{\text{stat}} = \text{PatchEmbed}(S) + E^{\text{2D}} \in \mathbb{R}^{B \times N_s \times D},$$

which pass through $\mathcal{T}_{\text{spatial}}$ and are mean-pooled:

$$s = \frac{1}{N_s} \sum_{i=1}^{N_s} U^{\text{stat}}_{:,i} \in \mathbb{R}^{B \times D}. \tag{5}$$

3.4.3. Feature-wise Linear Modulation (FiLM)

FiLM is a conditioning mechanism that uses a conditioning vector (in our case, derived from static basin attributes) to modulate the feature channels of another network via feature-wise affine transformations, as explained in Algorithm 2 and Figure 2 [62]. Formally, let $s \in \mathbb{R}^D$ denote the static embedding for a given basin, and let $z_t \in \mathbb{R}^D$ denote the dynamic embedding for day t. A small MLP (the "FiLM generator") computes:

$$(\gamma,\ \beta) = f_{\text{film}}(s), \qquad \gamma \in \mathbb{R}^D,\ \beta \in \mathbb{R}^D. \tag{6}$$

Each channel of z_t is then modulated by:

$$\tilde{z}_t = z_t\ \odot\ (1 + \gamma)\ +\ \beta, \tag{7}$$

where, $\odot$ denotes element-wise multiplication. This form (i.e., $1+\gamma$) ensures that at initialization the modulation is near the identity mapping, promoting stable training. By applying FiLM at each time step, the static context effectively shapes the dynamic embedding sequence to reflect basin-specific hydrologic response behaviours. For example, basins with high available water capacity may receive a γ that dampens the sensitivity to precipitation channels. In contrast, steeper basins may receive a γ that amplifies runoff-responsive features. In our architecture, this FiLM modulation is inserted after the spatial Transformer embedding and before the temporal Transformer encoder, thereby conditioning each day's representation on the basin's static attributes before modelling inter-day dependencies.

Algorithm 2 FiLM Conditioning of Dynamic Embeddings

Require: Static embedding $s \in \mathbb{R}^{B\times D}$, dynamic embedding sequence $\{z_t\}_{t=1}^{L} \subset \mathbb{R}^{B\times D}$
Ensure: Conditioned dynamic embeddings $\{\tilde{z}_t\}_{t=1}^{L} \subset \mathbb{R}^{B\times D}$
1: $(\gamma, \beta) \leftarrow f_{\text{film}}(s)$ ▷ MLP produces $\gamma, \beta \in \mathbb{R}^{B\times D}$
2: **for** $t = 1$ to L **do**
3: $\quad \tilde{z}_t \leftarrow z_t \odot (1+\gamma) + \beta$ ▷ feature-wise affine modulation
4: **end for**
5: **return** $\{\tilde{z}_1, \ldots, \tilde{z}_L\}$

3.4.4. Temporal encoding and streamflow prediction

The conditioned sequence is

$$\tilde{Z} = [\,\tilde{z}_1, \tilde{z}_2, \ldots, \tilde{z}_L] \in \mathbb{R}^{B\times L\times D}, \tag{8}$$

A 1-D sinusoidal positional encoding is added $E^{\text{1D}} \in \mathbb{R}^{L\times D}$:

$$\hat{Z} = \tilde{Z} + E^{\text{1D}}. \tag{9}$$

This $\hat{Z}$ passes through a temporal Transformer $\mathcal{T}_{\text{temporal}}$ (with optional causal mask):

$$H = \mathcal{T}_{\text{temporal}}(\hat{Z}) \in \mathbb{R}^{B\times L\times D}. \tag{10}$$

the final time-step representation is then extracted.

$$h = H_{:,\,L,\,:} \in \mathbb{R}^{B\times D}. \tag{11}$$

A simple regression head maps h into a discharge prediction:

$$\hat{y} = W_{\text{head}}\,\text{LayerNorm}(h) + b_{\text{head}}, \quad \hat{y} \in \mathbb{R}^{B\times 1}. \tag{12}$$

Equations (1)–(12) describe the full forward pass of the model: embedding each day's forcing grid, encoding static attributes, modulating dynamic representations via FiLM, capturing temporal dependencies via Transformer, and outputting a streamflow estimate. By treating static attributes as rasters and injecting them via FiLM rather than simple concatenation, our design retains within-basin spatial heterogeneity, enables basin-specific conditioning of dynamic responses, and leverages self-attention to capture multi-day hydrologic dependencies. Table A.1 shows the model architecture's layer layout.

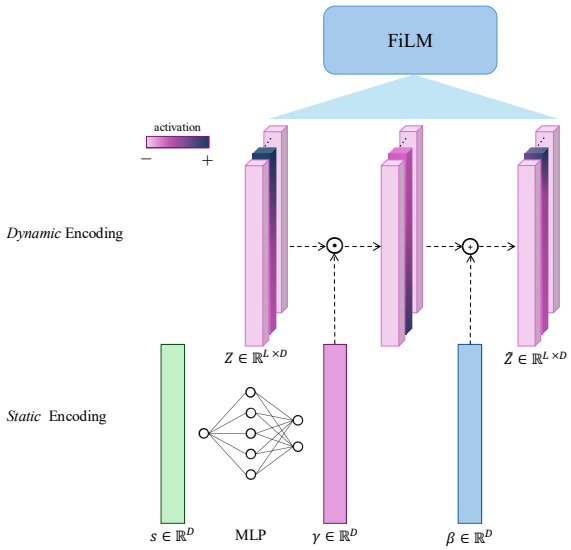


Figure 2: FiLM-based static-dynamic fusion: Static basin attributes are encoded through a multilayer perceptron to generate feature-wise modulation parameters ($\gamma, \beta \in \mathbb{R}^D$). These parameters are broadcast across all spatial–temporal dynamic patch embeddings Z and applied through an affine transformation $\tilde{Z} = \gamma \odot Z + \beta$. This mechanism enables static characteristics to influence how the model interprets spatial dynamic features.

3.5. Evaluation Metrics

To assess the predictive skill of the proposed model, two widely used hydrologic performance metrics are computed: the Nash–Sutcliffe Efficiency (NSE) and the Kling–Gupta Efficiency (KGE) [63, 64].

3.5.1. Nash–Sutcliffe Efficiency (NSE)

The NSE compares the variance of model residuals to the variance of the observed data and is defined as:

$$\mathrm{NSE} = 1 - \frac{\sum_{t=1}^{T} \left(Q_t^{\mathrm{obs}} - Q_t^{\mathrm{sim}}\right)^2}{\sum_{t=1}^{T} \left(Q_t^{\mathrm{obs}} - \overline{Q^{\mathrm{obs}}}\right)^2}, \tag{1}$$

where Q_t^{obs} and Q_t^{sim} denote observed and simulated daily discharge at time t, respectively, $\overline{Q^{\mathrm{obs}}}$ is the mean of the observed discharge series, and T is the number of time steps. A perfect model yields NSE = 1. An NSE of zero indicates that the model is as accurate as using the observed mean for

all predictions. Negative values imply that the observed mean is a better predictor than the model .

3.5.2. Kling–Gupta Efficiency (KGE)

The KGE was developed to overcome certain limitations of NSE by jointly accounting for correlation, bias, and variability of simulations relative to observations. It is defined as:

$$\mathrm{KGE} \;=\; 1 \;-\; \sqrt{\big(r-1\big)^2 + \big(\alpha-1\big)^2 + \big(\beta-1\big)^2}, \tag{2}$$

with

$$r = \text{Pearson correlation between } Q^{\mathrm{sim}} \text{ and } Q^{\mathrm{obs}}, \qquad \alpha = \frac{\sigma_{\,\mathrm{sim}}}{\sigma_{\,\mathrm{obs}}}, \qquad \beta = \frac{\mu_{\,\mathrm{sim}}}{\mu_{\,\mathrm{obs}}},$$

where, μ and σ denote the mean and standard deviation of the respective series. A value of KGE = 1 denotes perfect agreement; values range from -∞ to 1.

3.6. Experimental Setup

A series of controlled experiments was conducted to evaluate the proposed architecture, denoted as HydroTransformer, against four baseline models. These models include the standard LSTM, the EA-LSTM, a CNN-LSTM hybrid, and Informer. The prediction task is formulated as daily streamflow estimation, or nowcasting. For each target day, the models use meteorological input sequences of 5, 21, or 365 days. These sequences end on the target day to estimate streamflow for that same day. Therefore, the different sequence lengths represent different amounts of historical hydrometeorological context. They do not represent different future prediction horizons. These sequence lengths are used to evaluate the performance of the proposed FiLm-based static and dynamic fusion mechanism. This evaluation covers short-, medium-, and long-context settings. The 5-day setting emphasizes short-term storm response information. The 21-day setting captures intermediate antecedent moisture and recession behavior. The 365-day setting provides an annual scale, seasonal, and storage context. The experiments are designed to test the influence of antecedent input sequence length. They also test the use of static basin information. Finally, they test temporal and spatial generalization under holdout splits. See Table A.1 for the layer-level details of HydroTransformer.

The baseline models were selected to represent the main static–dynamic fusion paradigms used in deep-learning-based hydrologic modeling. Standard LSTM represents the common static variable concatenation and vector fusion paradigm. Entity-Aware Long Short-Term Memory represents the static gating paradigm. These two models cover the dominant ways static and dynamic variables have been fused in hydrologic deep learning. Convolutional Neural Network Long Short-Term Memory combines convolutional spatial feature extraction with recurrent temporal modeling. Informer is included as an additional Transformer-based sequence model. It provides an attention-based comparison against HydroTransformer. This allows us to assess whether the observed improvements are due only to Transformer-style temporal modeling or to the proposed spatially distributed static–dynamic fusion design.

All experiments were conducted on a server equipped with dual Intel Xeon Platinum 8568Y+ processors, with 48 cores per socket and two threads per core, 754 GiB of system memory, and NVIDIA L40S graphics processing units. The system ran Red Hat Enterprise Linux 9.4. The software environment used Python 3.13.12, PyTorch 2.10.0 with CUDA 12.8, and cuDNN 91002. HydroTransformer training was performed using four NVIDIA L40S graphics processing units in the distributed training configuration. Training time varied with sequence length and model configuration, ranging from approximately 5 hours for shorter sequence settings to approximately 18 hours for the longest 365-day sequence settings.

3.6.1. Sequence length experiments

The temporal context length $L \in \{5,\ 21,\ 365\}$ days is varied to assess how much antecedent information is required for each model to learn daily streamflow patterns. These three sequence lengths are chosen to represent distinct hydrologically meaningful time scales commonly studied in rainfall–runoff modeling.

A 5-day context captures short-term hydrologic responses driven by recent precipitation, evapotranspiration, and rapid runoff processes, and reflects the memory scale of fast-responding catchments. A 21-day context, approximately three weeks, is intended to capture intermediate-term basin memory, including soil moisture accumulation and delayed subsurface flow processes that influence streamflow beyond immediate rainfall events. Finally, a 365-day context represents long-term hydrologic memory, enabling the models to incorporate seasonal cycles, snow accumulation and melt, groundwater storage effects, and inter-annual climatic variability. This range of sequence

lengths allows us to assess each model's ability to leverage short-, medium-, and long-term dependencies in streamflow generation.

For each model, the input consists of dynamic meteorological forcings of length L, with static basin attributes included when supported by the model configuration, and the target is the same-day streamflow. For the LSTM and Informer baselines, static attributes are incorporated as vector-based auxiliary inputs when the static-input setting is enabled. For the EA–LSTM model, static basin attributes are required because they are used in the entity-aware gating mechanism. For the CNN–LSTM model, only dynamic meteorological inputs are used because the implemented architecture does not include a static-attribute fusion component. For HydroTransformer, the same sequence lengths are used, but the model is designed to ingest spatially distributed gridded forcings and to explicitly fuse them with distributed static basin rasters. This setup allows us to isolate the effects of temporal context length across architectures while also evaluating whether explicit distributed static–dynamic fusion improves daily streamflow estimation.

3.6.2. Spatial and temporal splitting strategy

The study adopts two complementary holdout schemes to assess temporal and spatial generalization.

Temporal split: Records from 2000–2013 are used for model training, while records from 2014–2020 are used for validation. All basins are included in both splits; thus, the temporal split tests how well models generalize to future years within the same watersheds.

Spatial split: Basins are partitioned into two sets: 47 watersheds for training, covering 2000–2013, and 12 distinct watersheds for validation, also covering 2000–2013. This spatial split ensures that the validation basins are never seen during training and thus quantifies the model's ability to generalize to new catchments, as shown in Figure 3.

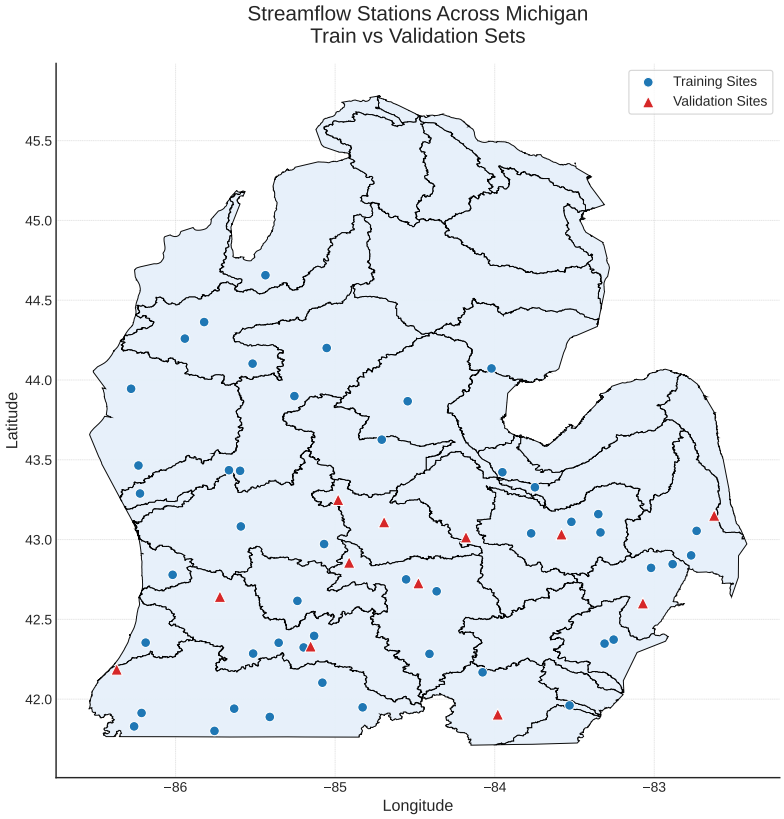


Figure 3: Spatial distribution of streamflow gauging stations across Michigan's Lower Peninsula, overlaid on watershed boundaries. Blue circles denote stations used for training, and red triangles indicate stations reserved for validation.

3.6.3. Model-specific considerations

The LSTM baseline represents the common concatenation or vector-fusion strategy for incorporating static attributes into sequence models. The EA–LSTM baseline is implemented only with static basin attributes and represents the static-gating paradigm. The CNN–LSTM baseline is implemented without static-attribute fusion, meaning only dynamic meteorological inputs are used. Informer is included as a Transformer-based sequence-modeling baseline and is evaluated with and without static vector inputs where applicable. HydroTransformer is designed to demonstrate the benefits of explicit spatial encoding and distributed static–dynamic fusion, using gridded meteorological forcings and static raster attributes rather than only basin-aggregated vectors. The overview of experimental configurations is presented

Model	Seq. Lengths	Static Input	Split Type
LSTM	{5, 21, 365}	Optional vector	Temporal & Spatial
EA–LSTM	{5, 21, 365}	Required vector	Temporal & Spatial
CNN–LSTM	{5, 21, 365}	None	Temporal & Spatial
Informer	{5, 21, 365}	Optional vector	Temporal & Spatial
HydroTransformer	{5, 21, 365}	Raster statics	Temporal & Spatial
Temporal Split: 61 watersheds; 2000–2013 training, 2014–2020 validation.			
Spatial Split: 47 training watersheds and 12 unseen validation watersheds.			

Table 4: Overview of experimental configurations used to evaluate baseline models and the proposed HydroTransformer.

in Table 4.

All models are evaluated using the same pre-processing pipeline, meteorological input variables, sequence lengths, temporal and spatial splits, and evaluation metrics. Hyperparameters are tuned using comparable validation procedures, with batch size adjusted only when required by model memory constraints. Performance is assessed using two hydrologic-standard metrics: Nash–Sutcliffe Efficiency (NSE) and Kling–Gupta Efficiency (KGE), computed over the validation sets.

To strengthen the experimental evaluation, we further perform statistical significance testing using basin-level model performance values and report average model rankings across metrics, sequence lengths, and evaluation splits. These analyses allow us to assess whether the observed differences are consistent across basins and settings rather than being driven only by aggregate mean or median values.

4. Results and Discussion

The performance of HydroTransformer is evaluated using (i) temporal holdout (future years in seen basins) and (ii) spatial holdout (ungauged basins). Each model is tested under identical meteorological inputs, static attributes (when supported), and sequence lengths [65]. Results are reported

using median and mean NSE/KGE across basins, and per-basin distributions are further analyzed to assess robustness. ***Note:*** For the LSTM [24] and EA-LSTM [25] experiments, 153 static variables are used because these models rely on aggregated inputs. In contrast, because processing spatial rasters is computationally expensive, only six spatial static attributes are retained for HydroTransformer.

4.1. Temporal Generalization

The temporal split experiment evaluates each model's ability to generalize to future hydrologic conditions within the same basins. Models are trained on the 2000–2013 period and tested on 2014–2016, ensuring that the evaluation reflects interannual climate variability, shifts in snow and rainfall regimes, and changes in antecedent soil moisture conditions rather than differences in spatial characteristics. This design isolates a model's temporal robustness, defined as its capacity to learn hydrologic response mechanisms that persist across hydrologically distinct years. Figure 4 presents the empirical cumulative distribution functions (CDFs) of NSE and KGE across all basins for sequence lengths of 5, 21, and 365 days with static variables. Notably, the HydroTransformer achieves consistently superior performance to all LSTM-based models while using just six static variables—far fewer than the 153 aggregated static inputs required by the LSTM benchmarks. For all sequence lengths and both metrics, the HydroTransformer curves lie distinctly to the right of the LSTM and EA–LSTM baselines, indicating superior across-basin performance. This consistent rightward shift signifies improved robustness across wet, dry, and transitional years in the validation period.

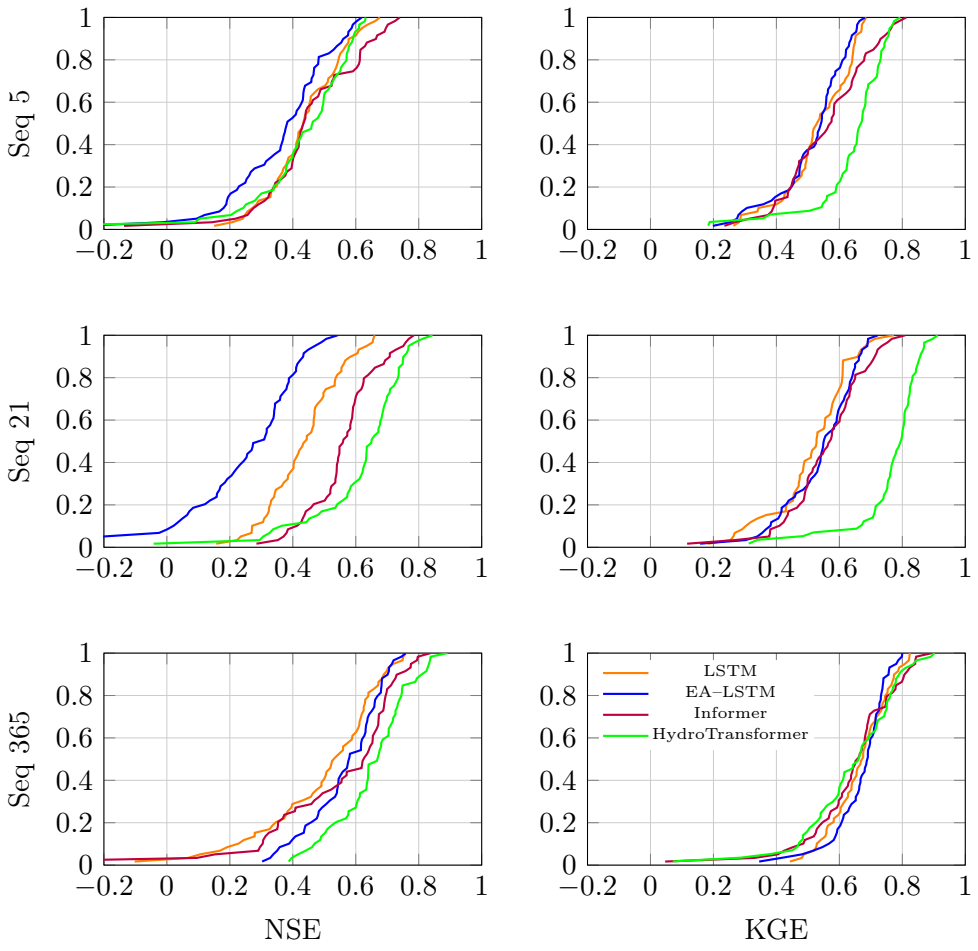


Figure 4: Comparison of temporal-split performance using empirical cumulative distribution functions of Nash–Sutcliffe Efficiency (left column) and Kling–Gupta Efficiency (right column) for sequence lengths of 5, 21, and 365 days (rows) using static variables. Curves show Long Short-Term Memory, Entity-Aware Long Short-Term Memory, Informer, and HydroTransformer with static inputs. A curve shifted further to the right indicates better performance across basins.

These CDF trends are corroborated quantitatively by the median NSE and KGE values summarized in Table 5. With static inputs enabled, the HydroTransformer achieves the highest median NSE across the short and medium sequence lengths (0.463 and 0.659 at 5 and 21 days, respectively),

and similarly leads in KGE (0.665 and 0.792). At the annual 365-day horizon, the HydroTransformer remains competitive but exhibits a slight reduction in relative advantage compared to the 21-day setting. This behavior is attributable to computational constraints, the 365-day configuration requires substantially larger input tensors, forcing the model to be trained with a significantly smaller batch size than the LSTM and CNN–LSTM baselines. Smaller batch sizes are known to produce higher gradient variance, slower convergence, and greater run-to-run instability. These effects are more pronounced in attention-heavy architectures. Consequently, the HydroTransformer's optimization at long sequence lengths is inherently noisier, which can modestly temper its final performance. Importantly, despite this limitation, the HydroTransformer still matches or exceeds the baselines at 365 days, and its superiority at 5- and 21-day windows indicates that when trained under comparable optimization conditions, it learns more stable and generalizable rainfall–runoff representations than all benchmark models.

The performance differences across sequence lengths also reveal important aspects of temporal information use. With short windows (5 days), all models capture only event-scale rainfall–runoff dynamics, yet HydroTransformer already demonstrates a clear advantage, suggesting that its spatially distributed forcings and FiLM-based conditioning allow it to represent storm-driven variability more effectively. At medium windows (21 days), which encode soil moisture memory, snowmelt transitions, and multi-storm interactions, performance gains widen. The HydroTransformer curve shows the strongest rightward displacement in both the NSE and KGE CDFs, and median scores peak at this horizon. This indicates that the attention-based temporal backbone is particularly effective at fusing medium-range dependencies that govern many basin-scale hydrologic processes. At long windows (365 days), the recurrent benchmarks exhibit mixed or diminishing returns, with LSTM and CNN–LSTM showing flatter or less stable CDFs, reflecting challenges in preserving long-term information within gated memory architectures. In contrast, the HydroTransformer maintains high median performance and favorable CDF shape, suggesting that the self-attention mechanism is better suited for extracting annual-scale signals such as snow accumulation–melt cycles, baseflow recession patterns, and seasonal storage changes.

Together, these findings demonstrate that the HydroTransformer provides substantially improved temporal generalization relative to traditional sequence models. Its ability to combine distributed daily forcings with basin-

	Temporal Split					
	Median NSE					
	Static: Without			Static: With		
Model	@5	@21	@365	@5	@21	@365
LSTM	0.180	0.063	0.119	0.439	0.444	0.535
EA–LSTM	–	–	–	0.383	0.309	0.579
CNN–LSTM	0.363	0.529	**0.639**	–	–	–
Informer	0.242	0.300	0.370	0.436	0.561	0.628
HydroTransformer	**0.458**	**0.658**	0.445	**0.463**	**0.659**	**0.670**
	Median KGE					
	Static: Without			Static: With		
LSTM	0.418	0.358	0.358	0.531	0.527	0.676
EA–LSTM	–	–	–	0.543	0.552	**0.691**
CNN–LSTM	0.637	0.701	**0.736**	–	–	–
Informer	0.380	0.430	0.572	0.575	0.573	0.662
HydroTransformer	**0.667**	**0.716**	0.629	**0.665**	**0.792**	0.658

Table 5: Median NSE and KGE of HydroTransformer and baseline models across sequence lengths {5, 21, 365} for the temporal split. Bold values indicate the best-performing model for each metric, static-input setting, and sequence length.

static conditioning, coupled with attention-based temporal modeling, enables it to learn hydrologic relationships that remain stable across years, hydrologic regimes, and sequence lengths. In Appendix, Figures A.1 and A.3 show the corresponding climatological hydrographs and observed–predicted scatter plots for the top-performing basins in the validation period.

4.2. Spatial Generalization for Ungauged basins

The spatial split experiment evaluates each model's ability to generalize to ungauged basins, where no discharge observations from the target basins are seen during training. Models are trained on a set of donor basins (set of basins for training) and evaluated on distinct basins (set of basins for validation) with different climatic, physiographic, and hydrologic characteristics.

TThis setup tests whether learned rainfall–runoff relationships can transfer across regions. This is an intrinsically more complex problem than temporal generalization, which only requires robustness to interannual variability within the same basin. Spatial generalization therefore depends critically on a model's ability to leverage static physiographic attributes and to interpret spatially distributed meteorological forcings, both of which strongly influence runoff generation in data-sparse environments.

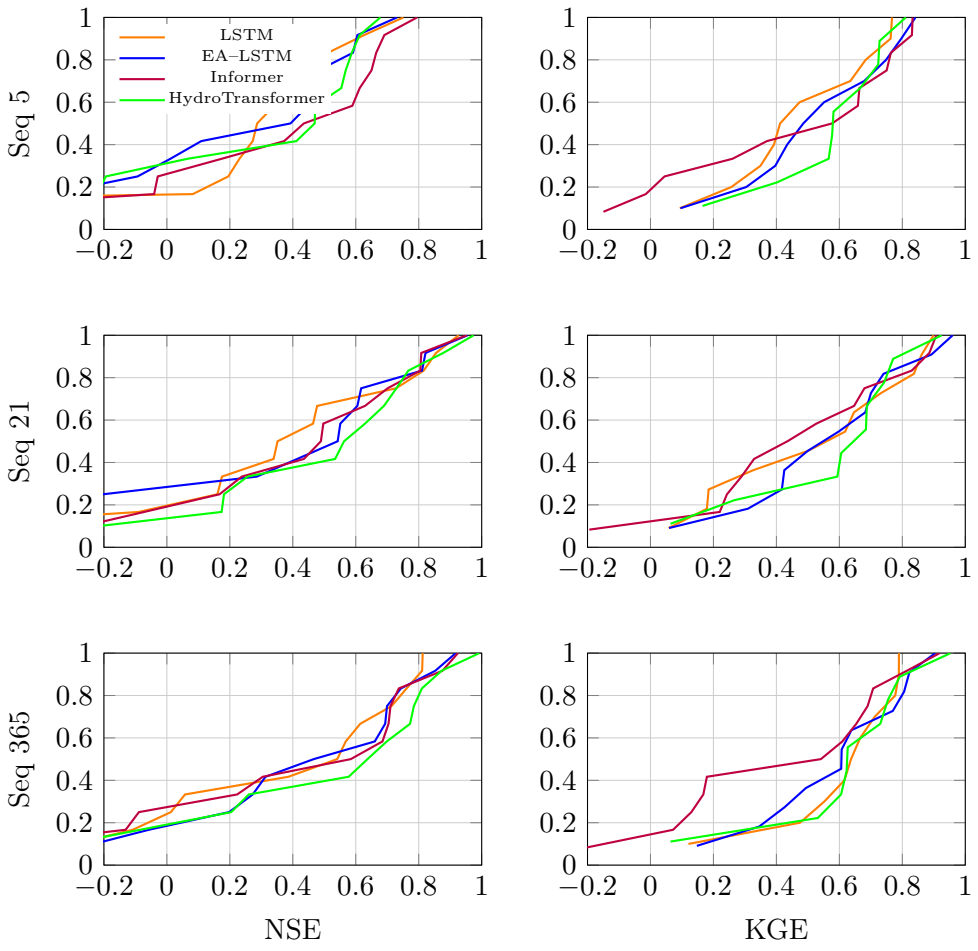


Figure 5: Comparison of spatial-split performance using empirical cumulative distribution functions of Nash–Sutcliffe Efficiency (left column) and Kling–Gupta Efficiency (right column) for sequence lengths of 5, 21, and 365 days (rows) using static variables. Curves show Long Short-Term Memory, Entity-Aware Long Short-Term Memory, Informer, and HydroTransformer with static inputs. A curve shifted further to the right indicates better performance across basins.

Figure 5 (the CDF curves for NSE and KGE) illustrates clear differences in how each architecture performs across ungauged basins. In the Static: Without setting, where models receive no physiographic information, performance is uniformly lower across all models. LSTM and CNN–LSTM achieve

moderate spatial skill, with median NSE values between 0.27–0.46 and mean NSE often below 0.35, consistent with the difficulty of transferring basin-aggregated temporal signatures to new hydrologic regimes. CNN–LSTM performs relatively well at shorter sequences (5–21 days), but its advantage diminishes at 365 days, suggesting limited cross-basin generalizability when relying solely on temporal feature extraction. HydroTransformer, despite its more expressive architecture, does not substantially outperform LSTM under the no-static condition, indicating that spatial representations alone are insufficient without basin-specific conditioning. When static attributes are included, the spatial CDFs in Figure 5 show a pronounced improvement for all models, but especially for HydroTransformer. The tabulated results in Table 6 confirm these trends quantitatively. Across all sequence lengths, HydroTransformer achieves the highest median NSE values (0.470, 0.596, and 0.668 for the 5-, 21-, and 365-day sequences) and the highest mean NSE values (0.286, 0.505, and 0.513). These gains reflect the model's ability to integrate distributed Daymet meteorological fields with basin-level physiographic attributes via FiLM conditioning, enabling it to learn hydrologic representations that transfer more robustly across diverse basins. Unlike LSTM-based models, which rely on aggregated inputs that weaken the link between meteorological variability and spatially structured runoff processes, HydroTransformer preserves fine-scale spatial patterns such as precipitation gradients and snowpack distribution, while simultaneously adapting to basin-level properties such as slope, soil texture, and drainage density.

The progression across sequence lengths provides additional insight. While LSTM and CNN–LSTM exhibit either diminishing returns or only modest improvements beyond a 21-day window, HydroTransformer continues to benefit from longer temporal context, achieving its strongest performance at the 365-day sequence. This suggests that cross-basin prediction requires models capable of leveraging long-range temporal dependencies, such as seasonal storage, snow accumulation and melt cycles, and evolving soil moisture memory. It also indicates that attention-based temporal integration is more effective than gated recurrent mechanisms for capturing basin-invariant seasonal signals. Overall, the combination of the spatial CDF behavior and the summary statistics in Table 6 demonstrates that HydroTransformer provides the most reliable and transferable performance across ungauged basins. Its ability to jointly encode distributed dynamic forcings and basin-specific static attributes allows it to generalize hydrologic processes more effectively than classical LSTM-based architectures, whether or not they incorporate static

	Spatial Split (Ungauged basins)					
	Median NSE					
	Static: Without			Static: With		
Model	@5	@21	@365	@5	@21	@365
LSTM	0.273	0.455	0.590	0.314	0.408	0.555
EA–LSTM	–	–	–	0.420	0.547	0.563
CNN–LSTM	0.295	0.444	0.595	–	–	–
Informer	**0.574**	**0.554**	**0.609**	**0.512**	0.493	0.634
HydroTransformer	0.304	0.439	0.510	0.470	**0.596**	**0.668**
	Mean NSE					
	Static: Without			Static: With		
LSTM	0.199	0.356	0.309	0.080	0.355	0.403
EA–LSTM	–	–	–	0.100	0.370	0.457
CNN–LSTM	0.170	0.297	0.296	–	–	–
Informer	**0.334**	**0.403**	**0.378**	**0.333**	0.445	0.408
HydroTransformer	0.146	0.267	0.316	0.286	**0.505**	**0.513**

Table 6: Median NSE and mean NSE for the spatial (ungauged-basin) split across sequence lengths {5, 21, 365}. Bold values indicate the best-performing model for each static-input setting and sequence length.

features through simplified gating or concatenation mechanisms. The Appendix provides additional visualizations, Figures A.2 and A.4 illustrate the climatological patterns and observed–predicted scatter plots, respectively, for the basins with highest median NSE values during the validation period. See detailed results Tables A.2 and A.3 for both temporal and spatial split under different experimental settings.

4.3. The Role of Static Variables in Hydrologic Generalization

To quantify the contribution of static basin attributes to HydroTransformer's skill, HT models trained *with* and *without* static inputs are compared across both temporal and spatial experiments. Figure 6 reports the change in median NSE (ΔNSE = $\text{NSE}_{\text{static}}$ – $\text{NSE}_{\text{nostatic}}$) for each sequence

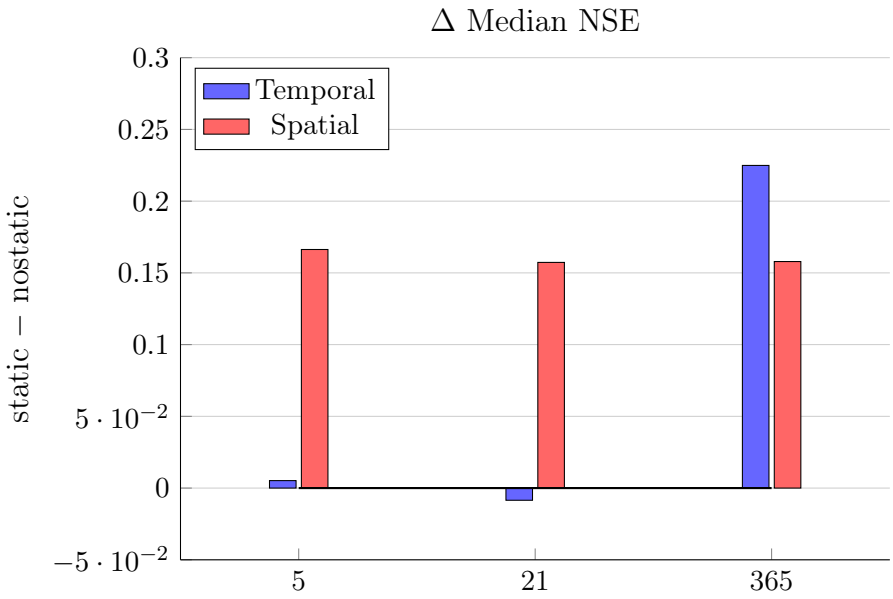


Figure 6: Change in median NSE (ΔNSE = $\text{NSE}_{\text{static}} - \text{NSE}_{\text{nostatic}}$) for HydroTransformer across temporal and spatial splits and sequence lengths {5, 21, 365}.

length. In the temporal split, static information provides negligible benefit for short sequences (5 and 21 days), but yields a substantial improvement for the 365-day configuration ($\Delta\text{NSE} \approx +0.22$). This suggests that static basin attributes become most valuable when the model has access to long antecedent histories, likely helping it interpret long-term hydrologic memory in a basin-specific manner rather than being redundant with the temporal context.

The pattern is markedly different in the spatial (ungauged) split. Across all sequence lengths, HydroTransformer benefits substantially from static variables, with ΔNSE values consistently around +0.15 to +0.17. The distribution of each watershed in Figure 7 confirm this trend: while temporal gains are centered near zero, spatial gains show a strong upward shift, indicating that static attributes improve performance for the majority of basins.

Overall, this ablation demonstrates that static physiographic information is essential for spatial generalization but only modestly beneficial for temporal extrapolation. When predicting in ungauged basins, where dynamic forcings alone are insufficient to infer basin structure, HydroTransformer relies heavily on static attributes to anchor its spatiotemporal representations and produce

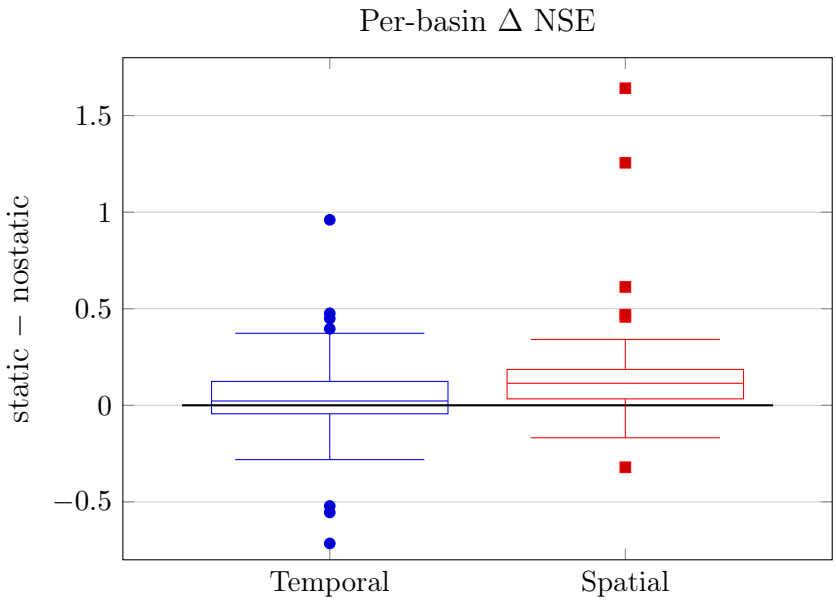


Figure 7: Per-basin differences in NSE ($\Delta\text{NSE}_i = \text{NSE}_{i,\text{static}} - \text{NSE}_{i,\text{nostatic}}$) for HydroTransformer, aggregated over all sequence lengths, for temporal and spatial splits.

reliable runoff estimates.

4.4. Effect of Sequence Length and Temporal Context

Figure 8 (temporal split) and Figure 9 (spatial split) summarize how mean NSE varies with input sequence length for LSTM, EA–LSTM, and HydroTransformer under static-input configurations. In both splits, model skill generally improves when moving from very short (5-day) to medium (21-day) context, reflecting the importance of capturing soil–moisture memory and multi–storm interactions. Beyond 21 days, however, the gains from extending the sequence to a full hydrologic year (365 days) are relatively modest in terms of mean NSE, especially compared to the additional computational cost of processing year-long spatial forcings.

This indicates that the Transformer's ability to exploit spatially distributed inputs and to perform attention-based temporal aggregation leads to a more efficient use of temporal context. A relatively short sequence (21 days) is sufficient for HydroTransformer to exceed the performance of recurrent baselines, even when those baselines use much longer and more computationally expensive 365-day sequences.

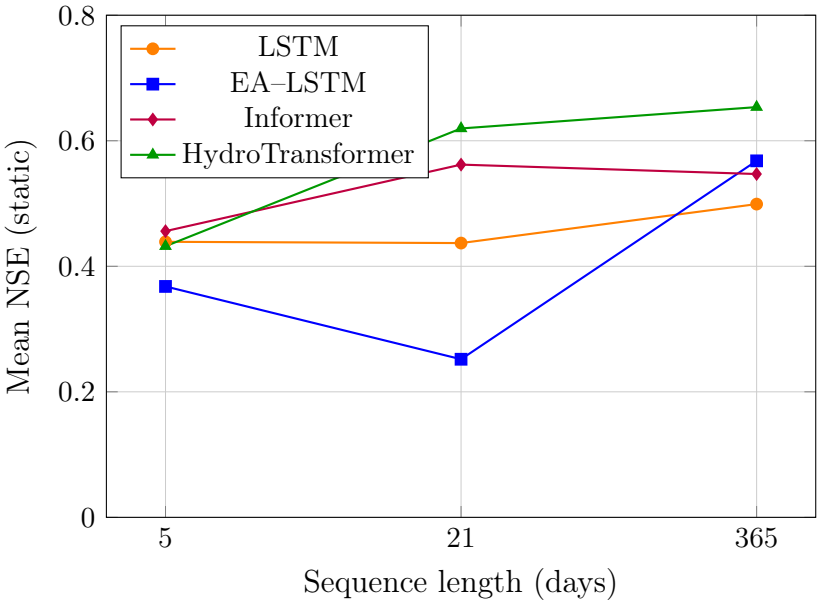


Figure 8: Effect of sequence length on mean NSE for the temporal split under static-input configurations.

4.5. Performance Across basin Areas

To examine how drainage area influences model skill, all basins are grouped into three physiographic classes (Small, Medium, Large) using terciles of basin area computed from the USGS attributes dataset. For each basin, NSE is averaged across all three temporal sequence lengths ({5, 21, 365}) under the static-input setting, and aggregated performance is computed for LSTM, EA-LSTM, and HydroTransformer.

Figure 10 summarizes results for the temporal split. Mean NSE increases from Small to Medium basins for all models, with a slight leveling off in the Large class. HydroTransformer achieves the highest score in every bin, with robust gains in the Small and Medium classes where hydrologic responses are typically flashier and less spatially averaged. This reflects the model's ability to leverage high-resolution dynamic inputs and static FiLM conditioning to stabilize predictions in physiographically challenging headwater basins. In contrast, LSTM and EA–LSTM show weaker and less consistent skill, particularly in Small basins. For the spatial split (Fig. 11), overall NSE values are lower—consistent with the difficulty of ungauged-basin prediction—and the performance gap between models narrows as sample sizes per class become

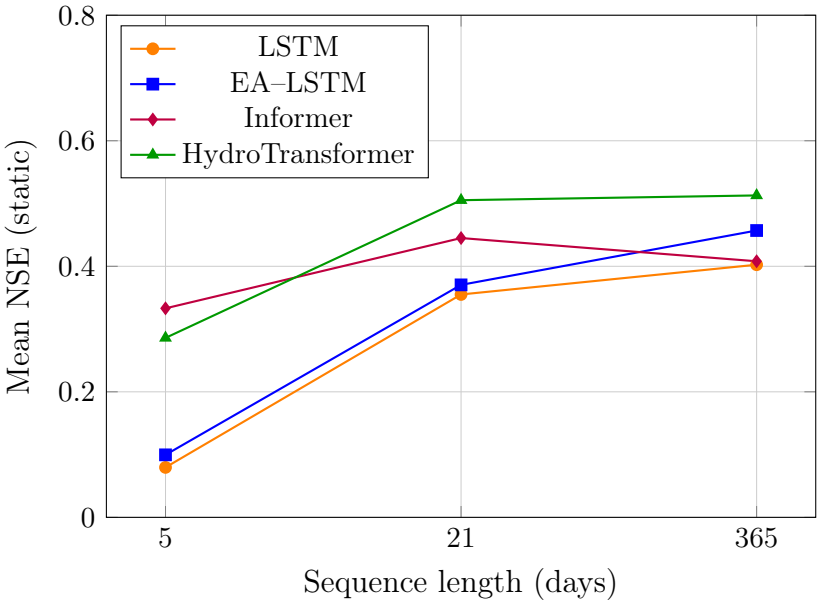


Figure 9: Effect of sequence length on mean NSE for the spatial split under static-input configurations.

smaller. Nevertheless, HydroTransformer maintains the top performance in Small and Medium basins and ties for the best in the Large class, indicating that its spatial encoding and conditioning mechanisms preserve robustness even when evaluating on entirely unseen basins.

Friedman Test Across Area Classes: To assess whether model differences within each area class are statistically significant, the Friedman test is applied to per-basin NSE values using the combined-sequence results ({5, 21, 365}). For the temporal split (Table 7), the null hypothesis of equal model performance is rejected in all three area classes ($p < 0.01$), with HydroTransformer achieving the lowest mean rank in every class. This confirms that the improvements observed in Fig. 10 are not only numerically higher but also statistically significant across Small, Medium, and Large basins. For the spatial split (Table 8), sample sizes per bin are much smaller, limiting statistical power. Accordingly, the Friedman test returns significant differences as the mean ranks continue to favor HydroTransformer in the Small and Large classes, in line with the aggregated trends in Fig. 11. While spatial generalization thus remains sample-limited, the ranking behavior suggests that Hydro-

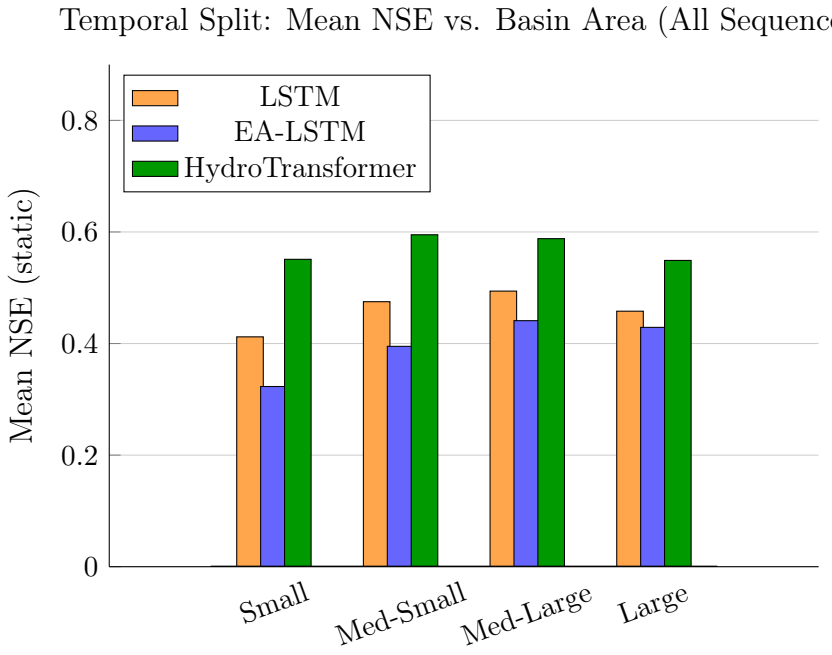


Figure 10: Temporal split: mean NSE across basin-area classes (Small, Medium, Large), aggregated over sequence lengths $\{5, 21, 365\}$ using static-input configurations.

Transformer still provides consistent advantages over LSTM-based baselines. Appendix B presents the results from the Friedman test.

Area class	n	χ^2_F	p	**Mean rank (1 = best)**		
				LSTM	EA–LSTM	HT
Small	18	31.00	1.86×10^{-7}	2.17	2.83	**1.00**
Medium	20	27.30	1.18×10^{-6}	1.95	2.85	**1.20**
Large	21	12.29	2.15×10^{-3}	2.24	2.38	**1.38**

* HT = HydroTransformer

Table 7: Friedman test across drainage-area classes for the temporal split, combining all sequence lengths ({5, 21, 365-days}) under static-input settings.

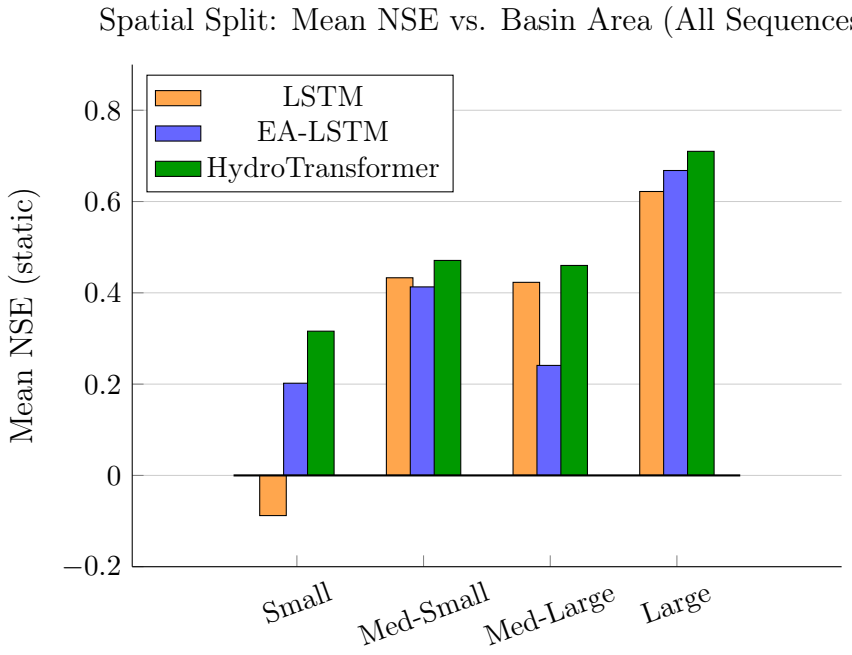


Figure 11: Spatial split: mean NSE across basin-area classes (Small, Medium, Large), aggregated over sequence lengths $\{5, 21, 365\}$ using static-input configurations.

Area class	n	χ^2_F	p	**Mean rank (1 = best)**		
				LSTM	EA–LSTM	HT
Small	5	0.40	0.819	2.20	2.00	**1.80**
Medium	4	0.00	1.000	2.00	2.00	2.00
Large	3	2.00	0.368	2.67	2.00	**1.67**

* HT = HydroTransformer

Table 8: Friedman test across drainage-area classes for the spatial (ungauged) split, combining all sequence lengths ({5, 21, 365-days}) under static-input settings.

4.6. Robustness of Model Performance Across basins

A complementary perspective to mean skill metrics is the robustness of each model across the full set of basins. Rather than relying solely on aggregated NSE values, each basin is categorized into four performance classes are High (NSE > 0.75), Moderate (0.36–0.75), Low (0–0.36), and Very Low

(≤ 0), which highlights how consistently a model performs across heterogeneous hydrologic settings. Tables 9 and 10 summarize these distributions for the 21-day sequence length with static attributes.

In the temporal split (Table 9), the HydroTransformer demonstrates markedly superior robustness relative to the LSTM and EA-LSTM baselines. It achieves the largest number of High-accuracy basins (7) and the smallest number of Low-accuracy basins (4). Traditional recurrent models, by contrast, show substantial concentration in the Low and Moderate categories, indicating greater sensitivity to basin-specific variability and interannual hydrologic fluctuations. The HydroTransformer's strong performance across nearly all basins underscores the benefit of integrating spatially distributed forcings with FiLM-based static conditioning. The spatial split (Table 10) presents a significantly more challenging generalization setting, yet HydroTransformer continues to demonstrate improved reliability. It maintains the lowest count of Negative-NSE basins (1) and achieves comparable or better representation in the High and Moderate categories compared to LSTM and EA-LSTM. While all models experience some degradation moving to the ungauged scenario, HydroTransformer's distribution remains the most balanced, indicating increased resilience to shifts in physiographic and climatic characteristics not observed during training.

Overall, the robustness analysis confirms that the HydroTransformer not only improves average accuracy but also reduces catastrophic failures, offering a more dependable predictive model across a diverse set of basins in both temporal and spatial generalization settings.

NSE Category	**LSTM**	**EA–LSTM**	**HydroTransformer**
High (>0.75)	0	0	**7**
Moderate (0.36–0.75)	43	18	**47**
Low (0–0.36)	16	37	**4**
Negative ($\leq$0)	0	4	**1**

Table 9: NSE robustness categories for the temporal split (21-day sequence, static inputs).

NSE Category	LSTM	EA–LSTM	HydroTransformer
High (>0.75)	3	3	**3**
Moderate (0.36–0.75)	3	5	**5**
Low (0–0.36)	4	1	**3**
Negative (≤0)	2	3	**1**

Table 10: NSE robustness categories for the spatial split (21-day sequence, static inputs).

4.7. FiLM Interpretability

To better understand the role of the Feature-wise Linear Modulation (FiLM) mechanism, an inference-time static-context perturbation analysis was conducted for HydroTransformer. The trained model was kept fixed, and only the static raster inputs provided to the FiLM branch were modified during evaluation. This design allows an examination of whether FiLM uses static basin information, whether the spatial organization of static rasters matters, and whether correct basin-specific static context contributes to prediction.

Three perturbation settings were considered. First, all static rasters were masked, removing the static conditioning signal provided to FiLM. Second, the spatial patterns of the static rasters were removed by replacing each raster with its basin-level spatial mean. This preserves the average value of each static variable while removing its within-basin spatial organization. Third, static rasters were shuffled across basins while keeping the dynamic meteorological inputs unchanged. This tests whether pairing meteorological forcing with the correct basin-specific static context is important.

Table 11 shows that perturbing the static context provided to the FiLM branch degrades HydroTransformer performance in both temporal and spatial split settings. In the temporal split, the baseline median NSE is 0.651. Masking all static rasters reduces median NSE to 0.386, while removing spatial patterns from the static rasters reduces median NSE to -0.128. Static shuffling has a smaller effect, reducing median NSE to 0.625. This suggests that, for basins represented during training, the model benefits from the spatial structure of static rasters. However, same-basin temporal prediction is less sensitive to basin identity mismatch because the dynamic meteorological sequence remains the dominant source of day-to-day streamflow variability.

Table 11: FiLM static-context perturbation analysis for HydroTransformer under temporal and spatial split settings. The trained model is kept fixed, and only the static raster inputs provided to the FiLM branch are perturbed during inference. Values shown are median Nash–Sutcliffe Efficiency (NSE) across basins.

Setting	**Temporal split**	**Spatial split**
Baseline: no perturbation	**0.651**	**0.474**
All static rasters masked	0.386	0.095
Static spatial patterns removed	−0.128	−0.262
Static rasters shuffled across basins	0.625	0.378

In the spatial split, where the model is evaluated on unseen basins, static-context perturbations also reduce performance. The baseline median NSE is 0.474. Masking all static rasters reduces median NSE to 0.095, and removing static spatial patterns reduces it to −0.262. Static shuffling also reduces median NSE to 0.378, indicating that correct basin-specific static context contributes to ungauged-basin transfer.

Overall, these results support the interpretation that FiLM acts as a distributed basin-conditioning mechanism. The strongest degradation occurs when the spatial patterns of static rasters are removed. This indicates that HydroTransformer benefits from spatially organized physiographic information rather than only basin-averaged static summaries.

4.8. Ablation Studies

To quantify the contribution of the main architectural components in HydroTransformer, an ablation study was conducted using the 21-day input sequence under both temporal and spatial split settings. The full HydroTransformer uses spatially distributed dynamic meteorological inputs, spatially distributed static basin rasters, and Feature-wise Linear Modulation (FiLM). FiLM is used to condition dynamic representations using static basin information.

The full model is compared against three controlled variants. The first variant removes FiLM conditioning. The second replaces spatially distributed dynamic meteorological rasters with basin-aggregated dynamic forcings. The third replaces spatially distributed static rasters with basin-aggregated static attributes. These variants are designed to evaluate whether the performance

Table 12: Architecture ablation study for HydroTransformer using the 21-day input sequence. All variants are trained and evaluated under the same temporal and spatial split settings. Values show median Nash–Sutcliffe Efficiency (NSE) across basins. "Full HydroTransformer" denotes the complete HydroTransformer; "Without FiLM conditioning" removes static–dynamic conditioning; "Aggregated dynamic inputs" replaces spatial dynamic rasters with basin-aggregated forcings; and "Aggregated static inputs" replaces spatial static rasters with basin-aggregated static attributes.

Model variant	**Temporal split**	**Spatial split**
Full HydroTransformer	**0.651**	**0.596**
Without FiLM conditioning	0.386	0.095
Aggregated dynamic inputs	0.423	0.221
Aggregated static inputs	0.432	0.223

gain comes from FiLM-based static–dynamic fusion, distributed meteorological inputs, and distributed static physiographic representation.

Table 12 shows that the full HydroTransformer achieves the highest median NSE in both the temporal and spatial splits. Removing FiLM conditioning causes the largest degradation, especially in the spatial split, where median NSE decreases from 0.596 to 0.095. This indicates that FiLM-based static–dynamic conditioning is central to the model's ability to represent basin-specific runoff response and transfer to unseen basins.

Replacing spatially distributed dynamic rasters with basin-aggregated forcings also reduces performance. The median NSE decreases to 0.423 in the temporal split and 0.221 in the spatial split. Similarly, replacing spatially distributed static rasters with basin-aggregated static attributes reduces median NSE to 0.432 and 0.223 for the temporal and spatial splits, respectively. These results show that preserving spatial structure in both meteorological forcings and static physiographic attributes improves streamflow estimation compared with aggregated representations.

The stronger degradation in the spatial split indicates that spatially explicit inputs and FiLM-based conditioning are particularly important for ungauged-basin transfer. Overall, the ablation results support the design of HydroTransformer as a distributed static–dynamic fusion architecture.

Controlled static-variable baseline experiment. To further isolate the effect of input representation, an additional controlled experiment was conducted.

The aggregated baseline models were trained on a reduced subset of static attributes. This subset corresponds to the physical variables used by the HydroTransformer. The selected attributes include elevation summaries and soil texture variables for clay, sand, and silt. Soil water content variables were used as the closest available proxies for available water content and field capacity. These two variables are not directly available in the aggregated attribute table. This experiment reduces the original 153 static attributes used by the LSTM baselines to a physiographically matched subset. This allows a more direct comparison between basin-averaged static descriptors and the spatially distributed static rasters of the HydroTransformer. Figure 13 compares the full static variable setting with the controlled static subset for LSTM, EA-LSTM, and Informer across temporal and spatial splits. The results show that restricting the baseline models to the matched static subset generally reduces performance. This reduction is especially notable for LSTM and EA-LSTM. In the temporal split, the LSTM median NSE decreases to 0.339, 0.346, and 0.334 for 5, 21, and 365-day sequences. The EA-LSTM median NSE decreases to 0.214, 0.240, and 0.309. In the spatial split, LSTM obtains median NSE values of 0.143, 0.381, and 0.399. EA-LSTM obtains 0.100, 0.533, and 0.560. Informer is less sensitive to the reduced static subset. It remains competitive, particularly for longer sequence lengths. These results indicate that the improvement of the HydroTransformer is not simply due to using a smaller or different set of static variables. The results instead support the value of preserving spatially distributed physiographic information. They also highlight the value of fusing this information with dynamic meteorological inputs through the proposed architecture. The reduced static baseline experiment complements the main comparison. It demonstrates that basin-averaged versions of the matched static information do not recover the performance obtained by the gridded static and dynamic representation of the HydroTransformer.

5. Conclusion

This study introduced **HydroTransformer**, a unified spatio-temporal Transformer architecture designed to improve daily streamflow estimation across heterogeneous and ungauged basins. By integrating distributed meteorological forcings with spatially explicit static attributes and fusing them via a FiLM-based modulation mechanism, the model captures basin-specific hydrologic behavior that traditional sequence models struggle to represent.

Extensive experiments across 59 watersheds demonstrated that HydroTransformer surpasses LSTM, EA–LSTM, and CNN–LSTM baselines in both temporal and spatial generalization settings, despite relying on far fewer static inputs. The temporal split results show that HydroTransformer achieves strong performance across a wide spectrum of hydrologic conditions, especially with medium-length (21-day) antecedent sequences where soil-moisture memory and multi-storm interactions dominate runoff generation. In the more challenging spatial split, HydroTransformer achieved substantial improvements in both median and mean NSE, confirming that distributed forcings combined with explicit static conditioning are essential for robust ungauged-basin prediction. Robustness analysis further revealed a reduction in low-skill and negative-NSE outcomes, indicating greater reliability across diverse hydrologic regimes.

Overall, this work highlights the value of combining high-resolution meteorological drivers with spatially distributed static basin attributes within a Transformer framework. Beyond improving accuracy, the architecture offers a scalable pathway toward regional and continental streamflow prediction without basin-specific calibration. Future research may incorporate river-network topology, hybrid physics-informed attention mechanisms, and large-sample continental datasets to advance generalizable hydrologic modeling.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Software and data availability

The source code for the various stages of this study is available in the repositories: https://github.com/Muhammad-Talha-MT/HydroTransformer.
The raw and processed data used for training, shape files, and interpretation can be provided on request.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT 5.2 to improve readability and language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Appendix A. HydroTransformer: Details and Performance

Block	**# Layers**	**Input dim**	**Output dim**
Dynamic input	–	$(B, L, C_{\text{dyn}}, H, W)$	$(B, L, C_{\text{dyn}}, H, W)$
Static input (resized)	–	$(B, C_{\text{stat}}, H_s, W_s)$	$(B, C_{\text{stat}}, H, W)$
Dynamic patch embed	1	$(B{\cdot}L, C_{\text{dyn}}, H, W)$	$(B{\cdot}L, N, D)$
Static patch embed	1	$(B, C_{\text{stat}}, H, W)$	(B, N, D)
Spatial encoder (dyn)	4	$(B{\cdot}L, N, D)$	$(B{\cdot}L, N, D)$
Mean pool (dyn)	1	$(B{\cdot}L, N, D)$	(B, L, D)
Spatial encoder (stat)	4	(B, N, D)	(B, N, D)
Mean pool + LN (stat)	1	(B, N, D)	(B, D)
FiLM MLP	1	(B, D)	$(B, 2D)$
FiLM conditioning	1	(B, L, D)	(B, L, D)
Temporal encoder	6	(B, L, D)	(B, L, D)
Temporal readout	1	(B, L, D)	(B, D)
Regression head	2	(B, D)	(B, H_{out})

Table A.1: Compact layer layout of HydroTransformer (fusion = `film`) with patch size P=8, embedding dimension D=192, Horizon Length H_{out}=1, 4-layer spatial encoder and 6-layer temporal encoder. $N = \frac{H}{P} \cdot \frac{W}{P}$.

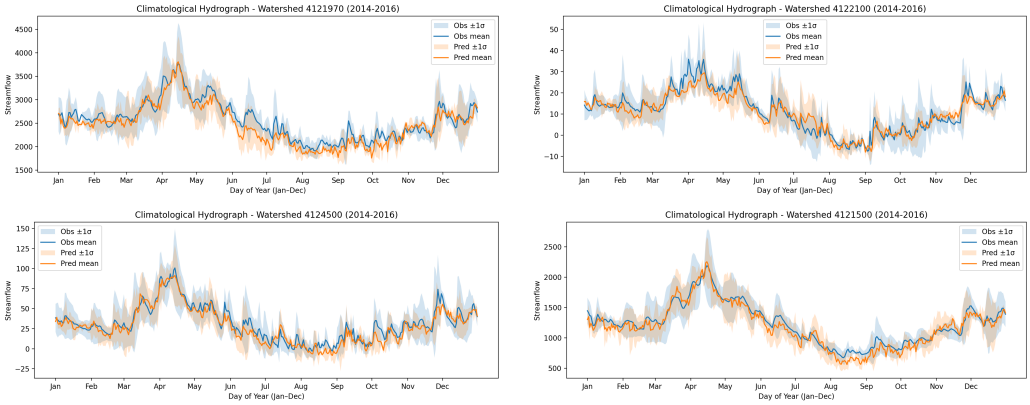


Figure A.1: *Temporal Split:* Climatological hydrograph for a representative watershed. Daily streamflow is aggregated by day of year over 2014–2016, with solid lines showing the mean observed (blue) and predicted (orange) streamflow and shaded regions denoting the $\pm 1\sigma$ spread across years.

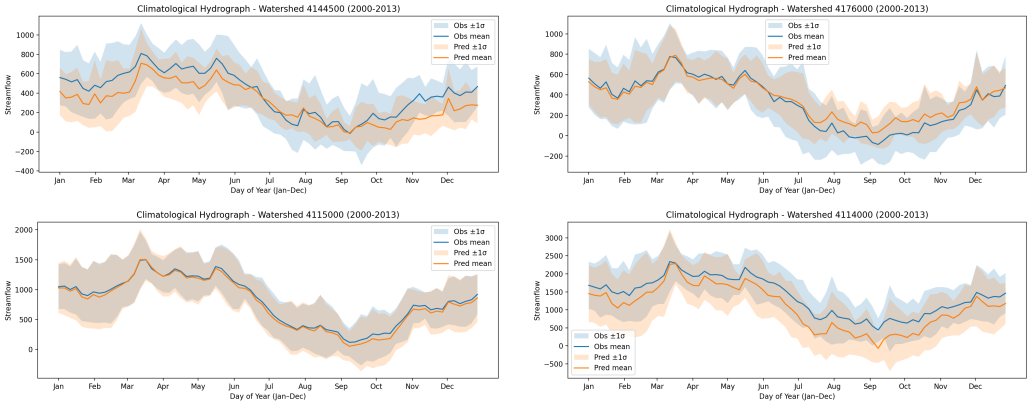


Figure A.2: *Spatial Split:* Climatological hydrograph for a representative watershed. Daily streamflow is aggregated by day of year over 2000–2013, with solid lines showing the mean observed (blue) and predicted (orange) streamflow and shaded regions denoting the $\pm 1\sigma$ spread across years.

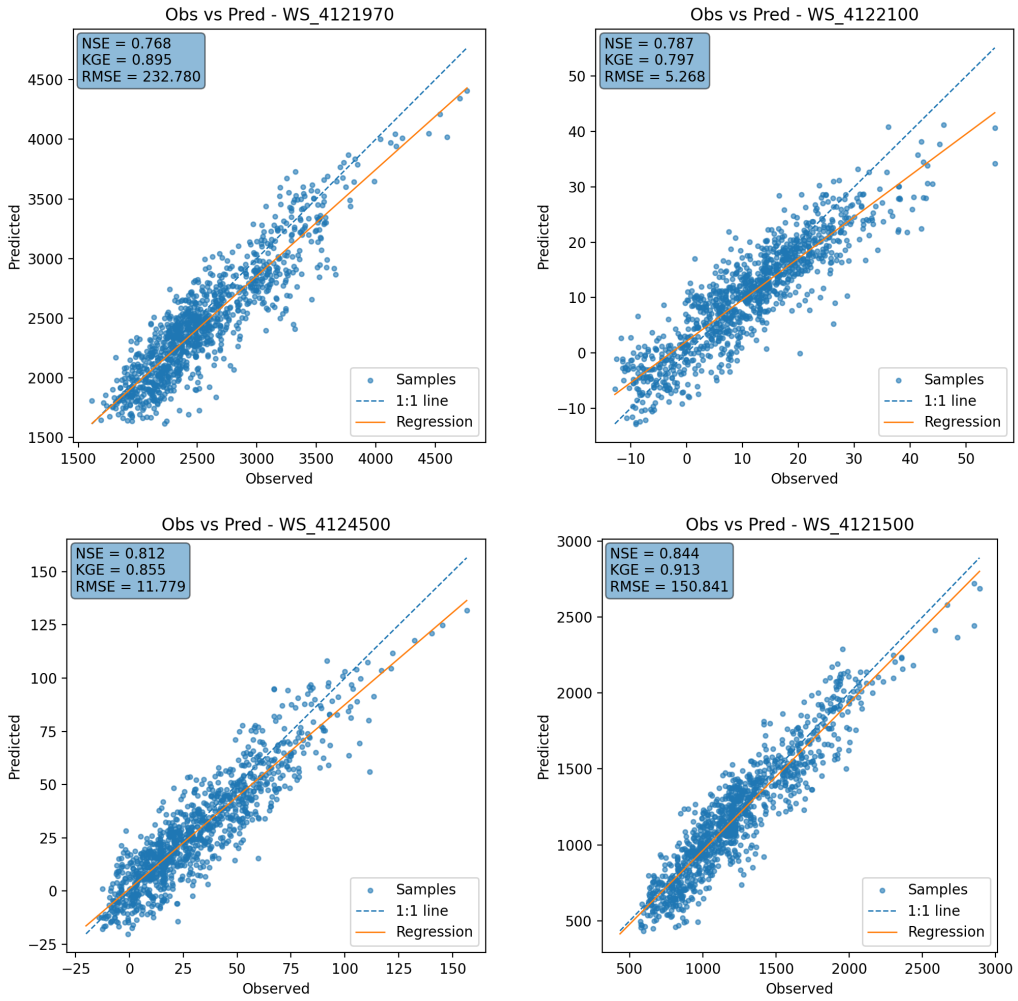


Figure A.3: Observed versus predicted streamflow for best watershed of temporal split. The scatter illustrates point-wise model performance, with the 1:1 line (dashed) indicating perfect agreement and the regression line (orange) showing systematic bias. NSE, KGE, and RMSE values summarize overall accuracy.

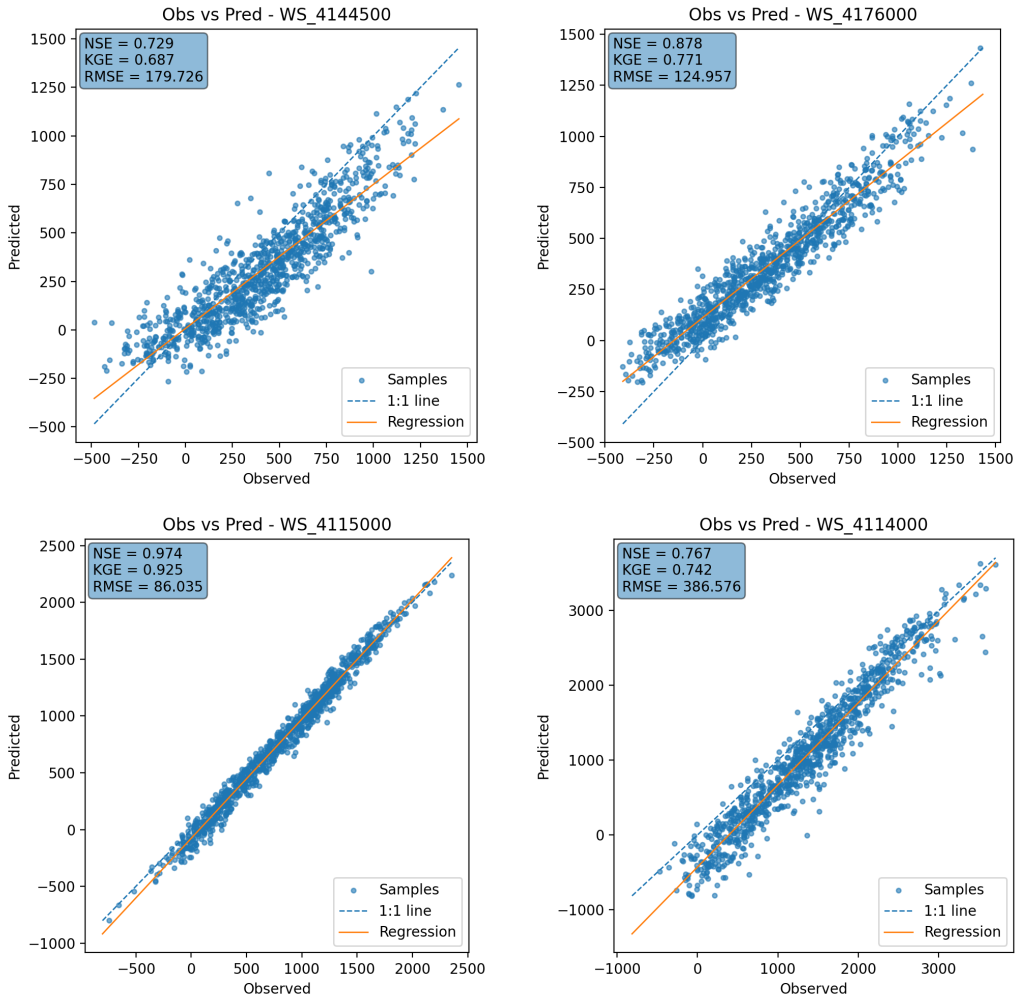


Figure A.4: Observed versus predicted streamflow for best watershed of spatial split. The scatter illustrates point-wise model performance, with the 1:1 line (dashed) indicating perfect agreement and the regression line (orange) showing systematic bias. NSE, KGE, and RMSE values summarize overall accuracy.

Table A.2: Temporal split: NSE/KGE summary statistics grouped by model, static setting, and sequence length.

Model	Static	Seq	Mean NSE	Median NSE/KGE	Min/Max NSE
EALSTM					
	Static				
		5	0.368	0.383 / 0.543	-0.270 / 0.619
		21	0.252	0.309 / 0.552	-0.343 / 0.543
		365	0.568	0.580 / 0.691	0.303 / 0.759
LSTM					
	Static				
		5	0.439	0.434 / 0.531	0.150 / 0.677
		21	0.437	0.444 / 0.527	0.157 / 0.660
		365	0.499	0.535 / 0.676	-0.102 / 0.759
	No Static				
		5	-0.224	0.193 / 0.418	-5.188 / 0.504
		21	-0.254	0.063 / 0.358	-4.941 / 0.408
		365	-0.083	0.119 / 0.358	-4.362 / 0.510
CNN-LSTM					
	No Static				
		5	0.214	0.363 / 0.637	-5.238 / 0.618
		21	0.425	0.529 / 0.701	-2.213 / 0.683
		365	0.549	0.640 / 0.736	-2.056 / 0.774
HydroTransformer					
	Static				
		5	0.432	0.464 / 0.665	-0.435 / 0.631
		21	0.620	0.651 / 0.793	-0.042 / 0.844
		365	0.654	0.670 / 0.658	0.387 / 0.891
	No Static				
		5	0.446	0.459 / 0.668	0.072 / 0.711
		21	0.624	0.659 / 0.717	-0.144 / 0.795
		365	0.387	0.445 / 0.629	-0.779 / 0.671

Table A.3: Spatial split: NSE/KGE summary statistics grouped by model, static setting, and sequence length.

Model	Static	Seq	Mean NSE	Median NSE / KGE	Min / Max NSE
EALSTM					
	Static				
		5	0.100	0.420 / 0.460	-2.170 / 0.731
		21	0.370	0.547 / 0.548	-0.617 / 0.955
		365	0.457	0.563 / 0.607	-0.275 / 0.918
LSTM					
	Static				
		5	0.086	0.338 / 0.479	-2.276 / 0.620
		21	0.281	0.478 / 0.551	-2.263 / 0.825
		365	0.403	0.555 / 0.627	-0.341 / 0.813
	No Static				
		21	0.309	0.590 / 0.600	-1.476 / 0.777
		365	0.243	0.477 / 0.514	-1.652 / 0.799
CNN-LSTM					
	No Static				
		5	0.231	0.387 / 0.569	-3.356 / 0.650
		21	0.328	0.503 / 0.592	-2.093 / 0.690
		365	0.316	0.510 / 0.531	-0.554 / 0.791
HydroTransformer					
	Static				
		5	0.365	0.500 / 0.623	-0.566 / 0.693
		21	0.472	0.571 / 0.681	-0.293 / 0.826
		365	0.550	0.565 / 0.589	0.131 / 0.811
	No Static				
		5	0.427	0.493 / 0.648	-0.690 / 0.714
		21	0.476	0.558 / 0.641	-0.462 / 0.781
		365	0.396	0.455 / 0.616	-1.228 / 0.718

Table A.4: HydroTransformer hydrograph-based diagnostic summary across sequence lengths and static-input settings. Values are medians across basins. Median RMSE is computed from the per-basin evaluation summaries, while high-flow RMSE, low-flow RMSE, and peak absolute error are computed from the simulated hydrographs. High-flow RMSE uses days above the basin-specific 95th percentile of observed discharge, low-flow RMSE uses days below the basin-specific 10th percentile, and peak absolute error is computed at the observed peak-flow day.

Setting	Seq.	Med. RMSE	High-flow RMSE	Low-flow RMSE	Peak AE
Temporal split ($n = 59$ basins)					
Without	5	75.1	130.9	60.8	82.3
Without	21	64.5	100.3	68.3	148.8
Without	365	75.7	181.9	61.8	273.0
With	5	87.9	147.2	61.9	171.3
With	21	68.1	111.0	51.6	138.4
With	365	54.9	112.4	51.6	117.1
Spatial split ($n = 12$ basins)					
Without	5	188.9	286.7	202.3	317.7
Without	21	144.9	168.2	166.0	269.8
Without	365	144.6	172.4	139.9	240.1
With	5	163.0	344.4	311.6	560.7
With	21	114.8	101.8	136.7	98.8
With	365	110.0	95.7	139.4	71.4

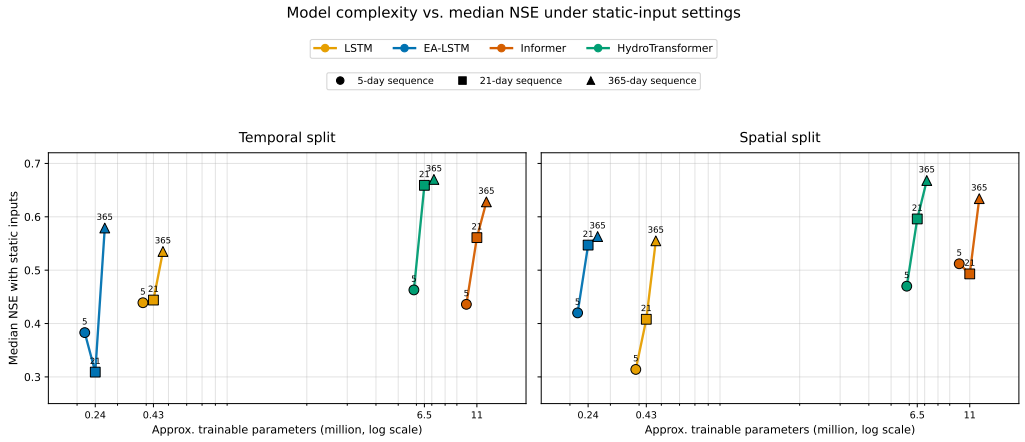


Figure A.6: Comparison of model complexity and predictive performance under static-input settings. The left panel shows temporal-split results and the right panel shows spatial-split results. The x-axis represents the approximate number of trainable parameters in millions on a logarithmic scale, and the y-axis shows median Nash–Sutcliffe Efficiency (NSE). Colors distinguish models, while marker shapes indicate sequence lengths of 5, 21, and 365 days. Lines connect the three sequence-length results for each model.

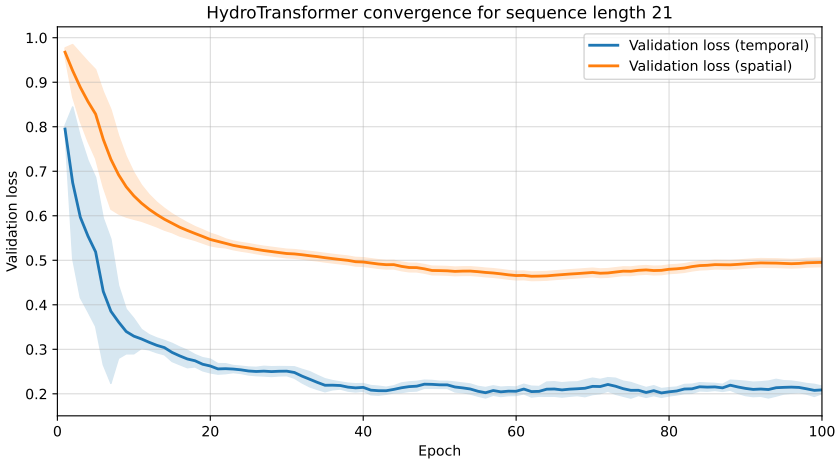
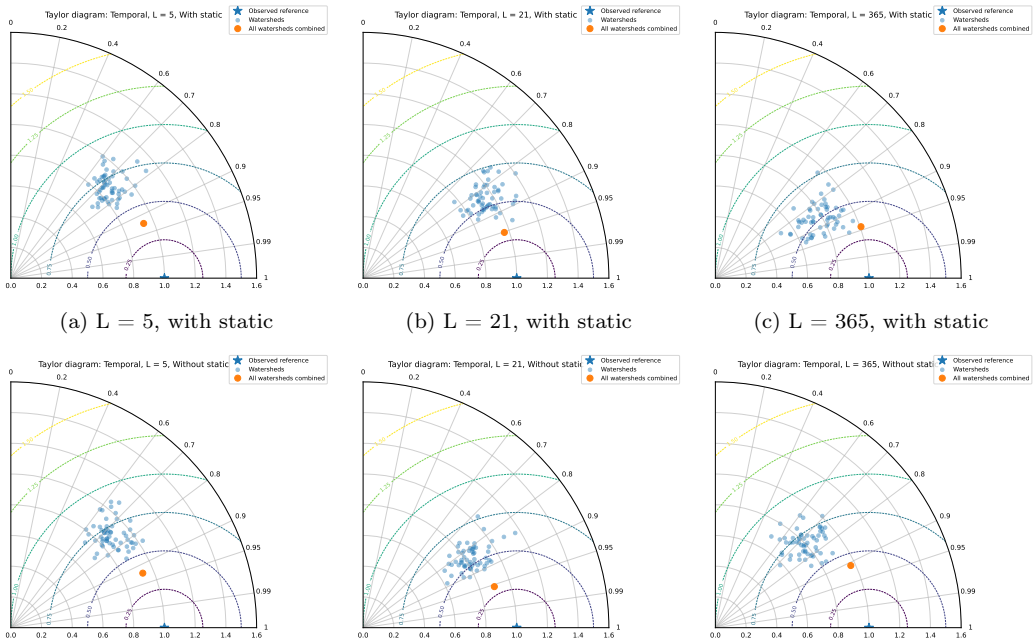


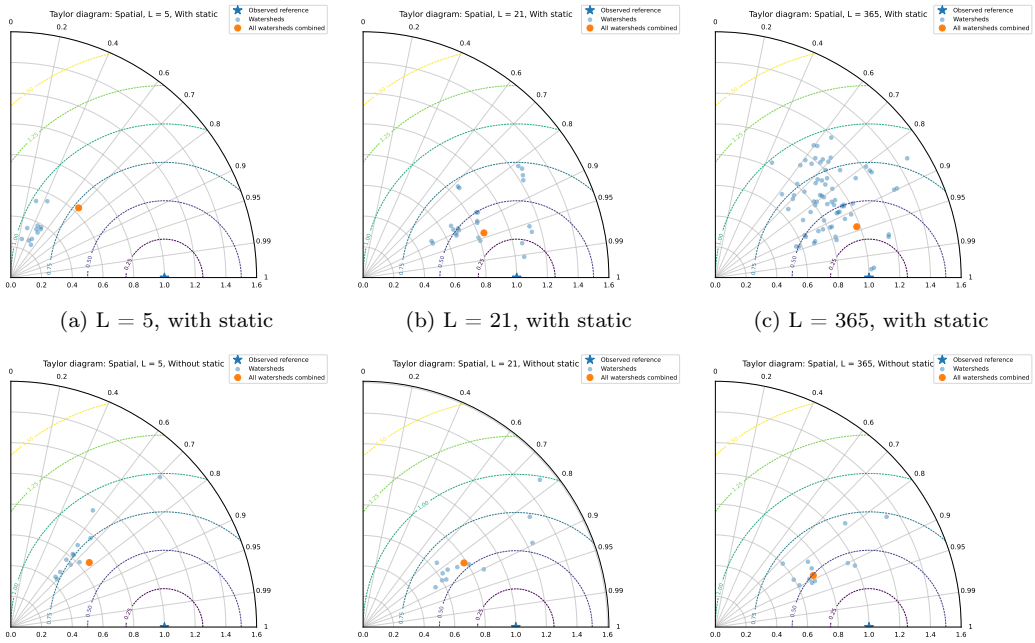
Figure A.5: Validation loss curves for HydroTransformer with a 21-day input sequence over the first 100 epochs under the temporal and spatial splits. Shaded bands indicate local variability around the smoothed validation-loss trajectory. The temporal split converges faster and stabilizes at a lower validation loss, whereas the spatial split stabilizes at a higher loss, reflecting the greater difficulty of ungauged-basin generalization.



(a) L = 5, with static (b) L = 21, with static (c) L = 365, with static

(d) L = 5, without static (e) L = 21, without static (f) L = 365, without static

Figure A.7: Taylor diagrams for HydroTransformer under the temporal split. Columns correspond to input sequence lengths of 5, 21, and 365 days, while rows compare settings with and without static basin attributes. Each point represents a watershed, and the combined marker summarizes performance across all watersheds. The diagrams jointly show correlation, normalized standard deviation, and normalized centered root-mean-square error.



(a) L = 5, with static (b) L = 21, with static (c) L = 365, with static

(d) L = 5, without static (e) L = 21, without static (f) L = 365, without static

Figure A.8: Taylor diagrams for HydroTransformer under the spatial ungauged-basin split. Columns correspond to input sequence lengths of 5, 21, and 365 days, while rows compare settings with and without static basin attributes. Each point represents a watershed, and the combined marker summarizes performance across all watersheds. The diagrams jointly show correlation, normalized standard deviation, and normalized centered root-mean-square error.

Appendix B. Statistical Analysis of Model Differences Across Watershed-Area Classes

This appendix presents a detailed statistical analysis evaluating whether the performance differences among LSTM, EA–LSTM, and HydroTransformer are significant within different watershed drainage-area classes. The Friedman test, a nonparametric alternative to repeated-measures ANOVA, is applied because it is appropriate for comparing multiple models evaluated within each area class across the same set of watersheds. The test assesses the null hypothesis:

H_0 : All three models have equal performance (same distribution of NSE).

For each watershed, NSE values were averaged across the three sequence lengths $\{5, 21, 365\}$ using only static-input configurations. The Friedman

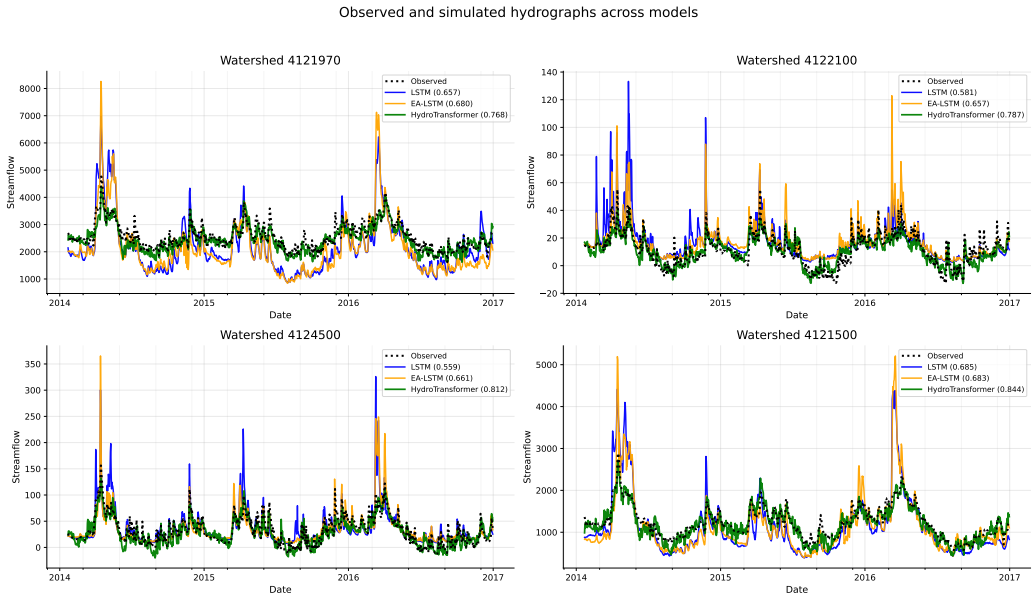


Figure A.9: Representative hydrograph comparisons for four watersheds under the temporal split using the 365-day static-input setting. Each panel shows the observed streamflow (black dotted line) together with simulated streamflow from Long Short-Term Memory (LSTM; blue), Entity-Aware Long Short-Term Memory (EA-LSTM; orange), and HydroTransformer (green). The figure highlights that HydroTransformer more closely tracks the observed hydrograph across diverse flow regimes, particularly in reproducing peak timing, recession behavior, and overall seasonal variability. Numbers in the legends indicate the Nash–Sutcliffe Efficiency (NSE) for each model over the displayed period.

test was conducted separately for the temporal and spatial splits and for each watershed area class (Small, Medium, Large), where area classes were determined using terciles of drainage area derived from the USGS attribute dataset.

For each class, the following are reported: (1) the number of watersheds n, (2) the Friedman test statistic χ_F^2, (3) the associated p-value, (4) the mean rank of each model (1 = best). Low p-values indicate significant differences in performance across models within that area class.

Split	Area Class	n	χ^2_F	p	Mean Rank (1 = best)		
					LSTM	EA–LSTM	HT
Temporal	Small	18	31.00	1.86×10^{-7}	2.17	2.83	**1.00**
	Medium	20	27.30	1.18×10^{-6}	1.95	2.85	**1.20**
	Large	21	12.29	2.15×10^{-3}	2.24	2.38	**1.38**
Spatial	Small	5	0.40	0.819	2.20	2.00	**1.80**
	Medium	4	0.00	1.000	2.00	2.00	2.00
	Large	3	2.00	0.368	2.67	2.00	**1.67**

* HT = HydroTransformer

Table B.5: Friedman test results across drainage-area classes for the temporal and spatial splits, using static-input experiments aggregated across all sequence lengths ($\{5, 21, 365\}$). Lower average ranks indicate better relative performance. Results for the spatial split should be interpreted with caution due to small sample sizes within each area class.

Appendix B.1. Interpretation

The temporal split exhibits strong statistical evidence that model differences are meaningful across all watershed-area classes ($p < 0.01$), with HydroTransformer achieving the best mean rank in every class. The magnitude of χ^2_F in the Small and Medium classes indicates particularly strong separation among models, reflecting HydroTransformer's advantage in capturing hydrologic responses in small and mid-size watersheds.

In the spatial split, statistical significance is not achieved in any area class. This outcome is expected due to the limited number of watersheds per class (3–5), which restricts the power of rank-based nonparametric tests. Nevertheless, mean ranks still favor HydroTransformer in the Small and Large classes, consistent with the qualitative improvements observed in the main text.

Overall, the results confirm that HydroTransformer's benefits are statistically robust in the temporal generalization setting, whereas limitations in watershed sample sizes prevent strong significance claims under the spatial (ungauged) scenario.